



**National University of Computer and Emerging Sciences, Lahore**




# Manzil: Intelligent Traffic & Emergency Routing


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## Authors' Declaration

We declare that the work presented in this Deliverable 2 report is our own project work and has not been submitted previously, in whole or in part, to another institution for an academic award. This document describes a proposed system and planned evaluation; it does not present unperformed experiments as completed results.

## Abstract

Manzil is a simulation-based research and development prototype for adaptive urban traffic control and emergency routing. It combines video-based vehicle detection and tracking, traffic-state estimation, comparative signal-control policies, congestion-aware ambulance routing, and a moving green corridor within a SUMO test environment. A React dashboard and FastAPI services are proposed to expose system state and experimental records. This report specifies the requirements, research method, architecture, and controlled evaluation protocol; implementation and measured results remain subsequent project work.

## Executive Summary

Traffic congestion affects daily movement, while delays to an ambulance can carry more serious consequences. Conventional signal plans are prepared for expected demand and cannot respond to every short-term imbalance. Route guidance based only on distance can also direct an emergency vehicle toward a road that is short but heavily queued. Manzil addresses these related problems through a controlled research prototype rather than a public-road deployment. It will examine how traffic observations can support responsive signal decisions, how an ambulance route can reflect current network conditions, and how selected intersections can provide temporary priority as the vehicle approaches. The expected value for decision makers is a repeatable environment in which alternative policies can be compared before any investment in roadside equipment or operational integration is considered. The project does not promise citywide improvement at this stage. It establishes a practical basis for determining whether an integrated approach is beneficial, what disruption it causes to ordinary traffic, and which limitations must be resolved before later field-oriented work could be justified.

The proposed prototype combines recorded-video traffic observation with a simulated urban road network. Recorded video will be used to assess whether vehicle detection and tracking can provide useful counts and queue-related indicators. SUMO will provide a safe and repeatable setting for comparing fixed-time, rule-based adaptive, and reinforcement-learning signal control. When an ambulance mission is created, the system will evaluate valid routes using travel-time and delay estimates rather than distance alone. A moving green corridor will prepare only the next relevant intersections, allow safe signal transitions, detect that the ambulance has passed, and return each intersection to normal operation. A web dashboard will present the current network, selected route, traffic conditions, corridor status, warnings, and experiment summaries. These capabilities are deliberately limited to a manageable prototype with one active ambulance. Manzil will not connect to government signals, consume live government CCTV, cover all of Lahore, perform enforcement, or claim the safety certification needed for operational emergency response.

Project success will be judged through transparent comparisons rather than a visually convincing demonstration. Normal-traffic experiments will replay matched demand scenarios under each signal controller and compare waiting time, queue length, travel time, throughput, and fairness between approaches. Emergency experiments will separate the benefit of congestion-aware routing from the additional benefit and disruption of signal priority. The evaluation will retain scenario definitions, random seeds, software versions, controller settings, route revisions, failures, and metric definitions so that a reported outcome can be reproduced. This Deliverable 2 report therefore concentrates on the project vision, research evidence, software requirements, proposed method, and system design. It reports no invented accuracy,

travel-time reduction, or completed implementation milestone. Several choices, including the exact Lahore network, final video sources, tracking configuration, reinforcement-learning algorithm, database engine, and hardware targets, remain subject to confirmation through implementation and baseline profiling. The immediate deliverable is a technically bounded plan that the team and evaluators can inspect before development results are presented.

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# Chapter 1 Introduction

Urban traffic control is a networked decision problem. Demand can change by approach, queue spillback at one junction can obstruct another, and a timing plan that is adequate in one period may waste capacity in another. Emergency travel adds a stricter timing objective. An ambulance does not benefit merely from a short geometric path if that path contains slow links, long queues, or signals that repeatedly stop it. These observations motivate Manzil, a research and development prototype that studies traffic perception, adaptive signal control, congestion-aware emergency routing, and coordinated signal priority in one simulation-based environment. Manzil is not presented as a deployable municipal controller. It is a controlled experimental platform based on the microscopic traffic simulator SUMO and its TraCI interface [1, 2]. The platform is intended to make policies repeatable, observable, and comparable before any real-world actuation is considered. Recorded video supplies a second, deliberately bounded input mode for studying whether vehicle detection and tracking can produce useful traffic indicators. The simulator remains the authoritative source of network state in routing and signal-control experiments, which prevents uncertain camera observations from being treated as exact network measurements.

## 1.1 Purpose of this Document

This Deliverable 2 report translates the approved proposal into a traceable research and software design. It defines the questions to be investigated, reviews the evidence behind major design choices, specifies externally visible requirements, and describes the methodology and architecture needed to conduct fair experiments. It also records the distinction between confirmed project scope, proposed technical choices, and future implementation work. The report does not claim experimental improvements, real-time performance, model accuracy, or deployment readiness. At the proposal stage, the problem definition, major functions, and conceptual architecture were established, while dataset preparation, implementation, tuning, and evaluation remained planned work. The methodology therefore defines how later claims must be produced rather than anticipating their outcome. This distinction is important because a successful simulation run or a plausible interface design does not demonstrate a measurable improvement. Any later performance statement must be tied to a recorded configuration, matched baseline, stated metric, and repeatable set of runs. In this way, the document serves both as the academic basis for assessment and as an implementation agreement that the team can use to prevent scope drift.

## 1.2 Intended Audience

The primary audience is the FAST–NUCES FYP supervisor and evaluation panel. They require enough detail to judge the relevance of the research questions, the feasibility of the proposed software, and the validity of the evaluation plan. The report is also an engineering reference for the student implementation team. Team members should be able to derive component responsibilities, interface fields, scenario configurations, fallback behavior, and planned verification from the chapters that follow. A secondary audience consists of prospective traffic-management and emergency-service stakeholders who may wish to understand the problem, operational assumptions, and limits of the prototype. These stakeholders are not represented as current clients, project partners, or approving authorities. The writing therefore explains domain terms when they first become important, but it retains the precision needed for implementation and academic review. Business-level implications are kept separate from technical evidence, and proposed functions are not described as operational capabilities. This combination allows different readers to locate the information relevant to them without weakening the distinction between an undergraduate research prototype and a real traffic-control system.

## 1.3 Project Status and Evidence Boundaries

Table 1.1 prevents proposed components from being read as completed ones. A choice is described as "proposed" where the repository contains a design decision but no implementation artifact or measured evidence. This convention is used throughout the report.

**Table 1.1: Project status at Deliverable 2**

| Area | Confirmed evidence | Not yet evidenced |
|---|---|---|
| Problem and scope | Approved proposal, objectives, exclusions, and simulation basis | Field deployment or agency integration |
| Architecture | Conceptual React/FastAPI/SUMO and perception decomposition | Deployed services and measured capacity |
| Traffic control | Fixed, rule-adaptive, and RL comparison plan | Final RL policy and comparative results |
| Emergency operation | Dynamic route and moving-corridor design | Validated timing parameters and achieved savings |
| Perception | YOLO-based detector and tracker direction | Final checkpoint, annotated split, and measured accuracy |
| Data persistence | Experiment records and candidate entities | Confirmed database engine and schema migrations |

Table 1.1 summarizes the evidence boundary applied throughout this report. The approved proposal confirms the problem, intended features, simulation basis, and broad technology direction, but the repository does not contain an implemented traffic controller, trained perception model, prepared road network, or experimental result set. Architecture diagrams therefore describe the system that is to be built, while performance requirements are stated as targets that must be measured later. The table also prevents a proposed database, tracker, or reinforcement-learning policy from becoming an accidental commitment merely because it is technically plausible. This status discipline affects the language used in every chapter. Requirements use "shall" to specify intended behavior, methodology sections use future-oriented descriptions for procedures, and the literature review reports findings only from cited studies. A later project report can update an item from proposed to implemented only when the corresponding code, configuration, test evidence, or experimental record is available. This approach preserves the credibility of Deliverable 2 and gives reviewers a clear basis for judging subsequent progress.

## 1.4 Definitions, Acronyms, and Abbreviations

**Table 1.2: Key terminology**

| Term | Definition in this report |
|---|---|
| A* | A shortest-path search that ranks candidates using accumulated cost plus an admissible lower-bound heuristic [3]. |
| API | Application Programming Interface. |
| CV | Computer vision; here, processing traffic video into observable vehicle-level and aggregate indicators. |
| Green corridor | A sequence of bounded, moving signal-priority interventions ahead of an ambulance; not permanent green on the whole route. |
| MOT | Multi-object tracking, which associates detections into identities across frames. |
| Queue proxy | An operational estimate based on tracked, slow or stationary vehicles in a defined lane region; not a direct physical queue measurement unless calibrated. |
| RL | Reinforcement learning, in which a policy is optimized using state, actions, and cumulative reward. |
| Signal preemption | Interruption or modification of normal signal operation to serve an emergency request, including transition and recovery. |
| SUMO | Simulation of Urban MObility, a microscopic road-traffic simulator. |
| TraCI | Traffic Control Interface, used by an external process to read and change a running SUMO simulation. |
| Traffic-state contract | Versioned representation of approach observations, signal state, source mode, timestamps, units, confidence, and validity. |
| Throughput | Vehicles completing a defined movement or leaving a measured network region per unit time. |
| Waiting time | Time accumulated under the explicitly configured low-speed threshold; the exact SUMO definition and threshold will be recorded with each experiment. |

Table 1.2 fixes the meaning of terms that could otherwise be interpreted differently across perception, simulation, and transportation research. In particular, a queue proxy derived from image regions is not presented as a calibrated physical queue unless the camera geometry and scale support that interpretation. Waiting time and throughput must also be connected to the exact simulator definition, sampling

boundary, and configured threshold used in an experiment. The distinction between signal priority and preemption matters because the corridor may interrupt normal control, while the term green corridor does not mean that every signal on a route remains green simultaneously. The traffic-state contract provides the common representation through which modules exchange observations, but it preserves source, time, unit, confidence, and validity information. These definitions are more than editorial conventions. They prevent measurement drift, constrain interface design, and determine which comparisons remain legitimate. Later chapters extend the same vocabulary rather than introducing competing names for the same road, movement, phase, route, or observation.

## 1.5 Report Organization

The remainder of the report follows the six-chapter structure required for Deliverable 2. Chapter 2 defines the problem domain, research questions, measurable objectives, scope boundaries, stakeholders, constraints, and criteria by which the project should be judged. Chapter 3 reviews research on vehicle perception, conventional and reinforcement-learning signal control, simulation, routing, and emergency preemption, then derives specific implications for Manzil. Chapter 4 expresses system behavior through functional and non-functional requirements, use cases, proposed interface wireframes, persistent data, risks, and traceability. Chapter 5 explains the two input modes, network preparation, perception pipeline, controller baselines, reinforcement-learning formulation, route costs, corridor policy, experimental comparisons, metrics, and threats to validity. Chapter 6 maps those decisions to a component architecture, domain model, interaction sequences, state transitions, data contracts, concurrency rules, fallback behavior, testing strategy, and maintainability policies. The appendices provide a requirement-coverage check and list information that must be confirmed before later reports make implementation or performance claims.

## 1.6 Conclusion

This chapter has established Manzil as a reproducible investigation of interacting traffic decisions rather than a collection of unrelated artificial-intelligence features. The system will study how observable traffic conditions can inform signal control, how a dynamic route can respond to network delay, and how an ambulance can receive temporary priority without ignoring the effect on ordinary traffic. It has also established the evidence boundary for the present deliverable. The proposal supports the intended scope and architecture, while implementation performance, accuracy, and deployment readiness remain unverified. The report is consequently written as a design and research plan whose later claims must be supported by recorded scenarios, baselines, configurations, and measurements. Defined terminology

ensures that video-derived proxies are not confused with exact simulator state and that a moving corridor is not reduced to an unsafe instruction to turn every signal green. These principles provide the basis for the project vision in Chapter 2, where the objectives, boundaries, stakeholders, and evaluation questions are developed in greater detail.

# Chapter 2 Project Vision

This chapter converts the broad motivation for Manzil into a bounded project vision that can guide both implementation and evaluation. It identifies the decisions that the prototype is expected to make, the evidence needed to judge those decisions, and the conditions under which any conclusion must remain limited to simulation and selected recorded video. The chapter begins with the relationship between normal traffic control and emergency movement, then states the problem in terms that can be investigated within an undergraduate project. Research questions separate perception quality, normal signal control, dynamic routing, corridor operation, and robustness so that an improvement in one area is not mistaken for success in every area. The objectives and scope retain the approved proposal while excluding forecasting, enforcement, citywide deployment, and direct operation of public signals. Constraints related to data, map quality, computing resources, and the simulation-to-reality gap are treated as design inputs rather than footnotes. Stakeholder expectations and success criteria then clarify who can use the prototype and what evidence would constitute a credible project outcome.

## 2.1 Problem Domain Overview

Traffic-signal control allocates limited green time among movements that cannot proceed together. Fixed plans are understandable, auditable, and easy to reproduce, but they cannot respond to every short-term difference in approach demand. Adaptive control can use measured queues and waiting time, yet a controller that improves the network average may still be unacceptable if it repeatedly postpones a lightly loaded approach. Coordination adds another layer because an action at one intersection changes arrivals and queues downstream. Emergency routing operates on the same network with a stricter travel-time objective. A route is useful only when its cost reflects current link travel time, queue discharge, signal delay, and any closure that makes a movement unavailable. Signal priority may reduce ambulance stops, but an unsafe transition or poorly managed recovery can transfer excessive delay to ordinary traffic. Manzil therefore treats normal control, route selection, priority activation, passage detection, and recovery as connected decisions. It retains separate measures for the ambulance and other vehicles so that emergency benefit cannot conceal network disruption.

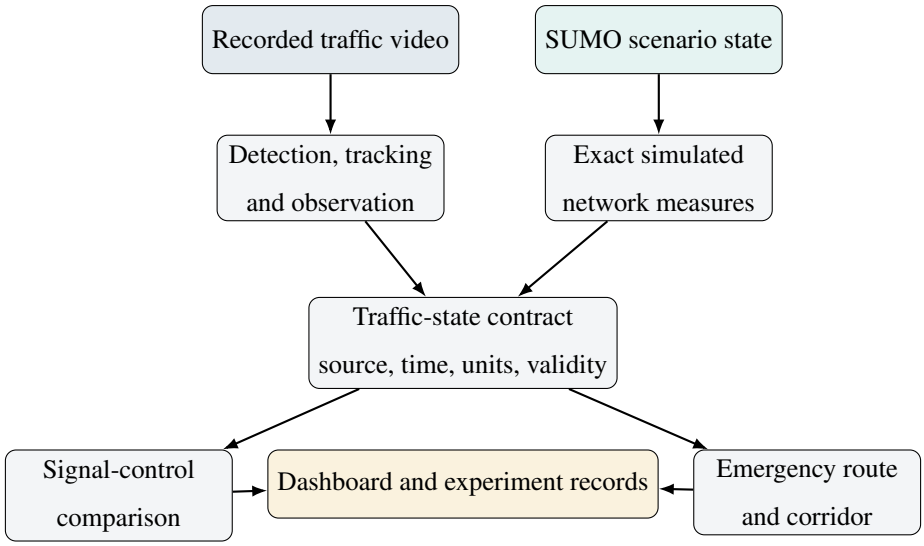



**Figure 2.1: Manzil project context and the separation of evidence paths**

Figure 2.1 presents the central scope decision that separates the two evidence paths while allowing them to use the same traffic-state contract. Recorded traffic video enters the perception path, where a detector, tracker, and region-assignment process produce observable measures that can be checked against annotations. SUMO enters the controlled-experiment path and exposes exact simulated vehicle, lane, route, and signal state for repeatable controller and routing comparisons. The shared contract records the source mode, identifier, timestamp, unit, confidence, and validity of every observation. This design permits the dashboard and downstream services to consume a consistent structure without implying that the sources have equal certainty. In particular, a count or queue proxy estimated from an arbitrary camera view cannot reconstruct every lane in the selected SUMO network unless camera calibration and lane mapping have been completed. The figure is relevant because it prevents a common scope error: presenting separate perception and simulation demonstrations as though they formed a validated real-time city model. Manzil will evaluate them separately unless a later, explicitly controlled integration study becomes feasible.

## 2.2 Problem Statement

Within the selected prototype setting, there is no established implementation that allows the Manzil team to compare signal-control strategies and isolate the contributions of emergency routing and coordinated priority under one reproducible protocol. The project problem is therefore to design a system that observes or simulates traffic state, makes bounded control and routing decisions, and records why each decision was requested and what was actually applied. The system must support matched comparisons between fixed-time, rule-based adaptive, and reinforcement-learning control under the same demand

conditions. It must also compare an unassisted ambulance, congestion-aware routing, and routing with a moving corridor without changing unrelated configuration between runs. This requires more than joining familiar technologies. Measurements must have defined units and freshness, routes must remain legal from the current vehicle position, signal changes must respect phase safety, and failed or infeasible runs must remain visible. The resulting evidence will describe performance in declared simulation and video conditions. It will not establish that the same effect would occur on public roads without calibration, controller integration, safety engineering, authority approval, and field validation.

## 2.3 Problem Elaboration

Five linked sub-problems determine whether the prototype can answer its research questions. First, frame-level vehicle detections must be associated over time and assigned to meaningful road regions before they can support counts, trajectories, or queue-related observations. Second, normal signal control requires a correctly configured fixed baseline and a transparent adaptive baseline before reinforcement learning can be judged fairly. Third, route costs must use compatible units and avoid adding separate congestion penalties when observed travel time already contains the same delay. Fourth, corridor control must anticipate the ambulance, clear conflicting movements, limit the duration of priority, detect passage, and restore normal service without holding the entire route green. Fifth, experiments must use matched demand, seeds, warm-up, duration, and metric definitions. These issues also interact across component boundaries. A stale observation can change an edge cost, a route revision can invalidate pending priority, and a prolonged priority phase can improve ambulance time while harming ordinary traffic. The vision consequently requires explicit freshness, single ownership of simulation time and actuation, safe fallback, route versioning, and auditable event order.

## 2.4 Research Questions

**RQ1** With declared video sources and annotations, how accurately and consistently can the proposed detector-tracker pipeline estimate observable vehicle counts and queue-related indicators?

**RQ2** Under matched SUMO demand scenarios, how do fixed-time, rule-based adaptive, and a selected RL controller differ in waiting time, queue length, travel time, throughput, and approach fairness?

**RQ3** Holding signal control constant, what change in ambulance travel time, stops, and route stability is attributable to dynamic congestion-aware routing?

**RQ4** Holding routing logic constant, what additional emergency benefit and ordinary-traffic disruption result from the moving green corridor?

**RQ5** How sensitive are the conclusions to demand level, incident placement, random seed, and observation uncertainty?

The research questions deliberately avoid the vague claim that Manzil will create "better traffic." Each question controls a specific factor and identifies evidence that can support or reject the corresponding claim. RQ1 concerns the reliability of observable video measurements and is judged against held-out annotations. RQ2 compares normal controllers under matched SUMO scenarios and includes fairness so that a lower network mean cannot hide starvation. RQ3 holds signal behavior constant while changing the route policy, which isolates the value of dynamic congestion information. RQ4 then holds routing logic constant while enabling the corridor, allowing its additional emergency benefit and its cost to ordinary traffic to be measured. RQ5 tests whether conclusions persist when demand, incidents, random seeds, or observation quality change. This separation also defines the order of implementation. The team can complete and validate fixed and rule-based control before training an RL policy, and it can validate route legality before adding priority. If one question cannot be answered within the available evidence, the remaining comparisons can still produce defensible findings without inventing an overall success claim.

## 2.5 Goals and Objectives

The goal is to develop and evaluate a simulation-based intelligent traffic-management prototype that supports normal traffic control and a single emergency mission. The measurable objectives are:

O1. Define and implement a versioned traffic-state representation shared by perception, simulation, control, routing, and visualization.

O2. Evaluate a pretrained vehicle detector with multi-object tracking on held-out, source-separated traffic-video segments.

O3. Construct a manageable multi-intersection SUMO network and repeatable demand scenarios.

O4. Implement fixed-time and reproducible rule-adaptive controllers, followed by one feasible RL controller.

O5. Compute valid ambulance routes using time-based dynamic edge costs and bounded rerouting.

O6. Coordinate safe, moving signal priority and restore normal control after passage or cancellation.

O7. Provide a React dashboard and FastAPI service boundary for scenario control, monitoring, and

experiment comparison.

O8. Record configurations and outcomes so each reported run can be reproduced and audited.

## 2.6 Project Scope

**Table 2.1: Scope boundaries**

| In scope | Outside the current scope |
|---|---|
| Recorded-video vehicle detection, tracking, counts, and queue proxies | Live government CCTV integration or citywide camera calibration |
| Selected multi-intersection SUMO network and synthetic demand | All of Lahore or safety-certified field deployment |
| Fixed, rule-adaptive, and one RL controller | Replacing real traffic-controller firmware |
| Single active ambulance with dynamic routing and moving priority | Multiple competing emergency fleets as a core case |
| Simulated closure or incident as a routing trigger | Visual accident classification or traffic forecasting |
| React dashboard, FastAPI services, and experiment records | Native mobile application, enforcement, facial or number-plate recognition |
| OpenStreetMap import with manual cleanup where feasible | A guarantee that map attributes are complete or current |

Table 2.1 separates the functions required for the research prototype from capabilities that would create unsupported operational or regulatory claims. The in-scope work is sufficient to examine the approved project idea: recorded-video traffic perception, a manageable multi-intersection simulation, three levels of normal signal control, congestion-aware routing for one ambulance, a moving priority corridor, and a dashboard with experimental records. The exclusions are equally important. Live government CCTV, direct operation of public signals, full-Lahore coverage, enforcement, number-plate or facial recognition, visual accident classification, forecasting, and multiple competing emergency fleets would require data, infrastructure, policy, and validation beyond the current project. A road closure may still be introduced as a controlled simulation event because it is needed to test rerouting, but the system will not claim to detect that incident visually. OpenStreetMap supports network preparation, yet its attributes will be checked and corrected rather than assumed complete. These boundaries keep implementation feasible and ensure that the final evaluation answers the stated research questions instead of dispersing effort

across unrelated features.

## 2.7 Sustainable Development Goals

Manzil aligns principally with United Nations Sustainable Development Goal 11, which concerns inclusive, safe, resilient, and sustainable cities [4]. The relevant connection is the investigation of how existing road and signal capacity might be used more responsively under changing traffic conditions, together with a structured method for prioritizing an emergency journey. The prototype can provide evidence about simulated waiting time, queue formation, throughput, ambulance travel time, and the recovery of ordinary traffic after priority. These measures are appropriate for a computing project because they can be defined and reproduced within the selected environment. The alignment must nevertheless be stated cautiously. A SUMO experiment does not demonstrate city-level sustainability, reduce emissions directly, or establish a public-health outcome. The project also does not model patient condition, clinical triage, hospital capacity, or the relationship between travel time and survival. Its secondary relevance to health is limited to studying a transport mechanism that may reduce simulated ambulance delay. Any stronger claim would require operational traffic data, emergency-service participation, field trials, and outcome measures outside the approved scope.

## 2.8 Constraints and Assumptions

The main constraints are the semester-scale implementation period, limited access to calibrated local traffic data, camera perspective and occlusion, finite model-training hardware, and the gap between simulated and real driving behavior. These constraints require a network small enough for repeated runs and detailed validation. A recognizable urban extract is preferred, but a documented synthetic grid remains an acceptable fallback if imported lane connections or signal programs cannot be corrected reliably within the schedule. The core experiments assume one active ambulance, valid origin and destination edges, controlled traffic generation, and no direct actuation of public infrastructure. Video-derived queue length remains an observable proxy unless the camera is calibrated to physical road geometry. The exact tracker, reinforcement-learning algorithm, database engine, hardware, map size, timing thresholds, and training budget are proposed choices rather than confirmed facts. These assumptions will be stored with experiment configurations because they affect interpretation. When an assumption fails, the system must report the failure or use a declared fallback instead of silently changing the scenario. This treatment makes constraints part of the research design and prevents later results from appearing more general than the tested conditions support.

## 2.9 Business and Practical Opportunity

The practical opportunity for Manzil is an experimentation and decision-support environment rather than an immediately deployable traffic product. A repeatable simulator can help a technical team compare how signal policies react to balanced, uneven, congested, and incident-affected demand. It can also show a prospective traffic or emergency-service stakeholder why the fastest ambulance route may differ from the shortest route, how a moving corridor progresses, and how much ordinary traffic is disrupted during and after priority. The dashboard and stored run manifests make those decisions visible instead of presenting the controller as a black box. This capability could support later feasibility studies, training demonstrations, or policy prototyping if the research results justify continued work. A commercial or public-sector deployment would require substantially more evidence: calibrated local demand, validated sensing, compatible roadside controllers, resilient communications, cybersecurity controls, privacy review, safety assurance, authority approval, and field trials. The current project assumes none of these partnerships or approvals and makes no revenue claim. Its immediate value lies in producing a defensible prototype and evaluation method that can reveal both benefits and limitations.

## 2.10 Stakeholders and User Characteristics

**Table 2.2: Stakeholder summary**

| Stakeholder | Interest | Expected interaction |
|---|---|---|
| Student research team | Correct implementation and defensible experiments | Configure, run, diagnose, export, and compare experiments |
| Supervisor and evaluators | Scope, rigor, traceability, and evidence quality | Review report, configurations, demonstrations, and results |
| Prototype operator | Clear scenario and emergency controls | Use the dashboard; inspect warnings and system state |
| Prospective traffic authority | Safety, reliability, network impact, auditability | Future consultation only; no current operational role |
| Prospective emergency service | Travel-time benefit and predictable priority behavior | Future requirements input only |
| Road users | Fairness and limited disruption | Indirect beneficiaries or affected parties in simulated evaluation |

Table 2.2 identifies the parties who influence the prototype or may later be affected by its decisions. The student team is the active system operator during development and evaluation, with responsibility for scenario configuration, execution, diagnosis, and evidence preservation. The supervisor and evaluation panel require transparent scope, reproducible methods, and a clear distinction between implemented and planned work. A prototype operator needs understandable controls and visible warnings but is not treated as an authorized emergency dispatcher. Traffic authorities and emergency services are prospective stakeholders whose operational knowledge would be necessary for any future deployment; they are not represented as current partners or as having approved the design. Road users are indirect affected parties because priority can shift delay between approaches even when an ambulance benefits. This stakeholder distinction shapes the interface and evaluation. The dashboard must expose source mode, stale data, fallback, route revisions, and corridor state to the operator, while experimental reporting must include ordinary-traffic effects for reviewers and prospective authorities. No stakeholder is assigned a role, credential, or commitment that is unsupported by the project record.

## 2.11 Success Criteria

Success is not defined by a visually impressive ambulance moving through green signals. The prototype must first behave correctly: scenario inputs must be validated, only one component may advance SUMO, signal transitions must remain legal, routes must follow permitted movements, obsolete corridor requests must be cancelled after rerouting, and failures must be recorded visibly. Research success then depends on comparison quality. Controllers must receive matched demand and seeds, emergency modes must differ only in the factor being tested, and metrics must retain the same definition across runs. Results should report variability rather than selecting a single favorable trace. A perception conclusion must refer to held-out annotations and the stated camera conditions, while a control conclusion must refer to the named network, demand family, run duration, and simulator configuration. Mixed or negative results remain valid if the protocol is followed. A method that improves ambulance time while causing unacceptable normal-traffic delay, starvation, unsafe transitions, or unstable rerouting cannot be described as successful overall. These criteria give the team concrete acceptance conditions and provide evaluators with a basis for judging both technical correctness and academic rigor.

## 2.12 Conclusion

This chapter has defined a project vision that is narrower than operational traffic management but broad enough to support a substantive research and development investigation. Manzil will connect traffic observation, comparative signal control, dynamic emergency routing, and a moving green corridor inside

a controlled and auditable prototype. The two evidence paths remain separate, and the core emergency case contains one active ambulance so that route and priority behavior can be examined without inventing an untested fleet-arbitration policy. Research questions identify the controlled comparison and corresponding measures for each claim. Scope boundaries exclude live government integration, city-wide deployment, forecasting, enforcement, and unsupported safety or health conclusions. Constraints concerning data, map quality, hardware, and simulation fidelity are incorporated into configuration and reporting. Stakeholder roles are limited to those supported by the project record, and success requires legal behavior, reproducible runs, visible failure, and balanced reporting of benefits and side effects. These decisions guide the literature review in Chapter 3, which examines the research basis for each technical component and identifies the limitations that Manzil's methodology must address.

# Chapter 3 Literature Review and Related Work

This chapter examines the research needed to justify Manzil's design rather than treating the use of familiar tools as a contribution. The review is organized around five decisions: how traffic is observed, how normal signals are controlled, how learned policies are evaluated, how emergency routes are selected, and how signal priority is activated and recovered. Each study is considered in its original data or simulation setting because reported values from incompatible benchmarks cannot be treated as a common comparison. The discussion identifies the method, evidence, limitations, and direct implication for a Manzil design choice. Research papers are distinguished from official software documentation, which is used to establish interface capabilities but is not counted toward the study requirement. The purpose is not to assemble every publication on intelligent transportation. It is to establish whether the proposed components are technically defensible, which assumptions must be recorded, and where a controlled undergraduate experiment can add useful evidence. The final synthesis consequently states a precise integration question instead of claiming that the use of YOLO, reinforcement learning, or shortest-path search is itself novel.

## 3.1 Review Method and Terminology

The approved proposal supplied the initial bibliography, after which each retained study was checked against a publisher page, proceedings record, DOI, or official open-access paper. Selection required direct relevance to vehicle detection and tracking, traffic-state estimation, signal control, simulation, graph routing, or emergency preemption. The review also applies consistent terminology. Detection refers to locating and classifying vehicles within one frame, while tracking refers to associating those observations through time. A detector therefore cannot produce reliable passage counts by itself. Signal priority describes preferential treatment that may remain within normal timing constraints, whereas preemption may interrupt normal operation for an emergency request. Manzil uses the term moving green corridor for a coordinated sequence of bounded interventions ahead of the ambulance, not for a permanently green route. These distinctions affect the interpretation of prior results and the design of the proposed experiment. A paper reporting benchmark detection accuracy does not establish a calibrated queue estimate, and a study of isolated preemption does not automatically validate network-level routing. The following sections preserve those boundaries when relating the literature to Manzil.

## 3.2 Vision-Based Traffic Observation

### 3.2.1 One-Stage Object Detection

Redmon et al. formulated YOLO as a single network that predicts bounding boxes and class probabilities from an image in one evaluation [5]. The study established the feasibility of real-time one-stage detection and reported a strong speed advantage, while also identifying localization error as a relative weakness. The design is relevant to Manzil because a compact inference path is practical for recorded traffic video and may later support a live demonstration on available hardware. The evidence does not establish accuracy for every vehicle class, road layout, camera height, weather condition, or Lahore traffic scene. A selected pretrained checkpoint must therefore be evaluated on the project's own held-out sequences, and confidence or intersection-over-union thresholds cannot be copied from an unrelated benchmark. Frame-level boxes also do not provide stable counts by themselves. The same vehicle may be detected in many frames, an occluded vehicle may disappear temporarily, and perspective changes apparent size across the image. Manzil consequently treats one-stage detection as the first part of a measurement pipeline that also requires tracking, road-region assignment, temporal aggregation, and explicit uncertainty.

### 3.2.2 Multi-Object Association

Wojke et al. extended a simple online tracker with a learned appearance metric to improve association through longer occlusions [6]. Appearance information can reduce identity switches, but it adds a feature-extraction model, processing cost, and sensitivity to domain differences. ByteTrack follows a different strategy by associating high-confidence detections first and then attempting to recover true objects from lower-confidence boxes rather than discarding them [7]. The authors evaluated it across several tracking benchmarks and reported strong accuracy with real-time throughput on their stated hardware. This approach is relevant to traffic video because occlusion and motion blur often reduce detection confidence. Neither benchmark establishes Manzil's processing speed, lane assignment, or queue accuracy. An identity-preserving track still requires a road-contact point, direction rule, and region geometry before it becomes a valid traffic observation. These studies therefore motivate an empirical tracker choice based on held-out count error, identity stability where identities are needed, failure under occlusion, and end-to-end throughput on the team's hardware. If a simpler association method produces equally reliable aggregate counts, its lower complexity may be preferable.

### 3.2.3 Traffic Datasets and Domain Coverage

BDD100K provides a large and diverse driving dataset with detection, tracking, and other driving-scene annotations collected across varied locations, weather, and times [8]. Its diversity is useful for model initialization and stress testing, although its primarily vehicle-mounted viewpoint differs from a fixed roadside or elevated camera. UA-DETRAC focuses directly on vehicle detection and tracking in traffic-surveillance video, making its camera geometry and evaluation protocol closer to an intersection-observation task [9]. Its scenes still cannot be assumed to represent Lahore traffic composition, motorcycle and rickshaw prevalence, lane discipline, lighting, or camera placement. The two datasets are therefore complementary references rather than automatic substitutes for local validation. Manzil should use licensed public sequences where they support reproducible evaluation and may add a small, separately documented local sample if collection permission and annotation effort are feasible. These datasets also influence the split strategy. Adjacent video frames are highly correlated, so a random frame split can place nearly identical observations in training and test data. The project will split by source or continuous sequence, retain a held-out evaluation set, and avoid tuning thresholds on that set.

## 3.3 Conventional and Adaptive Signal Control

### 3.3.1 Fixed-Time Baseline

A fixed-time controller remains necessary even when it is not expected to dominate every scenario. It provides a stable reference against which the value of traffic-responsive decisions can be measured, and it helps reveal whether demand generation or metric collection is biased before more complex controllers are introduced. The baseline must be credible rather than deliberately weak. Manzil will therefore document its phase sequence, green times, yellow and all-red clearance, cycle length, offsets where relevant, minimum service, and the demand conditions used to choose those settings. The same phase definitions and safety transitions will be shared with adaptive controllers so that a comparison does not reward one controller for using an illegal or more permissive movement set. Fixed timing is also useful for route and corridor ablations because its behavior is deterministic and easy to reproduce. If the fixed plan is severely under-saturated or over-saturated in every scenario, differences in controller performance will be difficult to interpret. A basic capacity and queue check will therefore confirm that the scenario family contains meaningful light, medium, and heavy conditions instead of a single demand level chosen to favor adaptation.

### 3.3.2 Max-Pressure Control

Varaiya formalized max-pressure control for a network of signalized intersections using differences between upstream and downstream queue conditions [10]. Selecting a phase that reduces network pressure has a clearer transportation basis than combining arbitrary percentages without explaining what the score represents. The approach is responsive and can operate from local movement information, which makes it a valuable non-learning reference for Manzil. Its assumptions still matter. Queue observations must correspond to valid movements, downstream storage must be represented, and a theoretical selection rule does not remove practical minimum-green, yellow, all-red, or maximum-red constraints. Manzil's initial rule-based adaptive controller will therefore use a pressure-inspired score only when the required upstream and downstream measures are available and sufficiently fresh. Waiting time or maximum-red information will remain as an explicit fairness safeguard so that a low-volume approach is not postponed indefinitely. The controller will also provide a reason for each request, allowing reviewers to see which queues and constraints influenced the decision. This transparency makes the baseline useful even if the later reinforcement-learning policy does not outperform it.

## 3.4 Reinforcement Learning for Traffic Signals

### 3.4.1 Cycle-Level Deep Q-Learning

Liang et al. used deep reinforcement learning to adjust traffic-light cycle behavior from observed traffic state [11]. The study demonstrates how an action definition, decision interval, state representation, and reward can be joined into a trainable signal controller. It also illustrates a limitation common to traffic RL: performance obtained for a particular intersection, simulator configuration, demand pattern, and training budget cannot be assumed to transfer to a different network. Manzil will therefore not reproduce a reported improvement as an expected project target or select deep Q-learning merely because it is familiar. A value-based method is feasible only after the discrete action space is bounded, legal phase transitions are enforced outside the learned policy, and fixed and rule-based baselines behave correctly. State normalization and reward components must be declared before evaluation, while training and tuning scenarios must remain separate from held-out tests. The policy will be assessed using traffic measures and safety interventions, not training return alone. If a simpler controller performs as well under the selected scenarios, that result will be reported rather than interpreted as a failure to produce a sufficiently complex model.

### 3.4.2 Pressure-Informed Reward Design

PressLight connects reinforcement-learning state and reward design to the transportation concept of max pressure [12]. The paper argues that a theoretically grounded pressure signal can reduce reliance on arbitrary handcrafted objectives and evaluates the method against conventional and learning baselines in arterial networks. This is relevant to Manzil because a reward based only on total waiting time may conceal spillback, unbalanced approaches, or repeated postponement of a minor movement. Pressure can provide a principal network-efficiency signal, while explicit penalties or constraints can address starvation and unnecessary switching. The study does not remove the need for careful evaluation. Topology, movement mapping, flows, phase definitions, observation range, and training budget all influence the learned result. Manzil must therefore record the exact state features, normalization, action timing, reward components, and safety interventions used by the selected policy. Emergency priority will initially remain a supervisory state machine rather than a rare reward event folded into normal training. That separation permits the normal-controller comparison to be interpreted independently and lets the additional effect of the corridor be measured through a controlled ablation.

### 3.4.3 Phase Competition and Network Cooperation

FRAP represents competition among signal phases and exploits structural relationships so that equivalent movements need not be learned as unrelated cases [13]. This is useful where an intersection contains several competing phases, but it requires an accurate phase graph and a more specialized representation. CoLight addresses cooperation across intersections through graph attention, allowing neighboring conditions to influence an agent's decision [14]. Chu et al. similarly study decentralized multi-agent deep reinforcement learning at larger scale and emphasize the importance of communication and partial observability [15]. Together, these studies show why the phrase "multi-intersection RL" is not a complete method. A project must state whether one centralized policy chooses all actions, independent agents operate without communication, or cooperating agents exchange defined observations. Multi-agent learning also introduces non-stationarity because each agent experiences an environment affected by policies that are changing at the same time. For Manzil, attention-based coordination may exceed the available tuning budget and make attribution difficult. The initial design will therefore use a small network and one explicit ownership model, adopting useful phase-structure ideas without claiming to reproduce every cited architecture.

### 3.4.4 Simulation Environments

CityFlow was introduced as a reinforcement-learning environment designed for large-scale city traffic and high simulation throughput [16]. It is an important alternative when training speed and very large networks dominate the design. SUMO is retained for Manzil because the approved proposal already depends on its microscopic vehicle model, traffic-light programs, OpenStreetMap conversion tools, NetEdit workflow, and TraCI interface. Lopez et al. describe the simulator's architecture and its support for detailed, externally controlled experiments [1]. This combination is appropriate for a manageable multi-intersection prototype in which route validity, individual ambulance movement, signal transitions, and per-vehicle outcomes must be inspected. Retaining SUMO also avoids the additional work of showing that a migrated network and demand model behave equivalently in another simulator. The choice does not make simulation results equivalent to field evidence. Car-following parameters, lane connections, signal programs, vehicle insertion, and map cleanup can materially change outcomes. Manzil will therefore version the network, demand, routes, simulator settings, and software release, and it will limit conclusions to the declared simulation conditions.

## 3.5 Dynamic Routing

Dijkstra's algorithm finds a minimum-cost path on a graph whose edge weights are nonnegative [17]. Hart, Nilsson, and Raphael established the formal basis of A*, which adds a heuristic lower-bound estimate to guide the search [3]. Both methods are well established, so selecting either algorithm is not a research contribution by itself. Manzil's project-specific task is to build a graph that respects valid SUMO vehicle movements and to define a cost that reflects time rather than distance alone. The proposed cost combines link travel time with non-overlapping estimates of queue discharge and residual signal delay, all expressed in seconds. If measured link travel time already includes queue delay, a separate congestion term cannot be added without double-counting. Dijkstra will provide the reference route, while A* is permitted only when free-flow travel time forms an admissible lower bound under nonnegative added delay. Dynamic rerouting also needs stability. A route will change when the current path becomes infeasible or when an alternative offers a persistent improvement above a defined threshold. Route revisions, no-route cases, and computation time will be evaluated rather than treated as incidental log messages.

## 3.6 Emergency Routing and Signal Preemption

Humagain et al. systematically reviewed emergency-vehicle route optimization and preemption methods and identified recurring concerns involving dynamic traffic data, travel-time objectives, ordinary-traffic effects, safety, and limited real-world application [18]. The review supports an integrated evaluation but does not justify the unsupported claim that integrated solutions do not exist. Qin and Khan examine transitions into emergency preemption and recovery to normal timing, including clearance and the queues created by interruption [19]. Their work shows that preemption is a lifecycle rather than a single instruction to set an approach green. Louati et al. investigate queue-aware emergency crossing across signalized intersections [20]. This is relevant because a green signal does not guarantee an unobstructed approach when the standing queue cannot discharge before the emergency vehicle arrives. Manzil therefore models requested, safe-transition, active-priority, passage, recovery, and cancellation states. It also uses look-ahead activation, timeout, downstream checks, and bounded priority duration. The precise timing values remain configurable simulation policies, not certified real-world safety parameters, and the evaluation will report ordinary-traffic disruption together with ambulance benefit.

## 3.7 Synthesis and Research Gap

The literature establishes reliable building blocks while exposing the risks that arise when they are connected. One-stage detection is fast enough to investigate traffic video, but useful traffic indicators require tracking, road geometry, temporal rules, and validation on the intended viewpoint. Max-pressure and reinforcement-learning research provide responsive control strategies, yet their comparison depends on state representation, phase constraints, reward design, training data, and simulator configuration. Dijkstra and A* solve well-defined path problems, but emergency routing remains only as credible as its movement graph, time-based costs, observation freshness, and rerouting stability. Preemption research makes safe transition and recovery central, and the review of emergency methods shows why ordinary-traffic disruption cannot be omitted. Manzil's proposed contribution is consequently not a new detector, shortest-path algorithm, or reinforcement-learning family. It is the controlled evaluation of an integrated and traceable prototype in which perception and simulation evidence remain distinct, normal controllers receive matched scenarios, routing and corridor effects are separated, and ambulance benefit is reported with network disruption and recovery. The unresolved question is whether that integration yields consistent benefits under the selected scenarios and constraints.

Table 3.1 consolidates the evidence discussed in this chapter so that the basis for each design choice can be checked without treating unlike experiments as a numerical league table. The second column

records the method and the setting in which evidence was obtained, while the remaining columns separate limitations from the decision that Manzil draws from the study. This distinction is important because benchmark accuracy, simulator throughput, and travel-time improvements measure different constructs and cannot be ranked on one common scale. The table also makes the review auditable: each row is tied to a cited source, each limitation identifies a boundary that the project must respect, and each implication points forward to a requirement, design element, or evaluation control. Portrait presentation is retained by combining the method and evidence into one wrapped column instead of rotating the page. The resulting summary is deliberately concise; detailed qualifications remain in the preceding thematic discussion, where datasets, assumptions, and transfer limitations can be explained in context.

**Table 3.1: Literature review summary**

| Study | Method and evidence | Limitations for Manzil | Design implication |
|---|---|---|---|
| Redmon et al. [5] | Unified one-stage detection; demonstrated real-time operation and a useful speed–accuracy tradeoff on standard benchmarks. | Original classes and benchmarks do not establish roadside traffic accuracy. | Evaluate a pretrained descendant on selected traffic video. |
| Wojke et al. [6] | Online tracking with learned appearance association; improved association through occlusion in its pedestrian setting. | Adds computation and a domain-dependent appearance model. | Consider it when identity fragmentation affects counts. |
| Zhang et al. [7] | Two-stage association that retains low-score boxes; strong results on several tracking benchmarks. | Benchmark identities and speed do not imply calibrated traffic measures. | Measure end-to-end count error before selecting the tracker. |
| Yu et al. [8] | Large multi-task driving-video dataset with varied conditions and annotations. | Mostly vehicle-mounted viewpoints differ from roadside Lahore scenes. | Use for initialization or stress tests, not as sole validation. |
| Wen et al. [9] | Vehicle detection and tracking benchmark for surveillance scenes with a joint evaluation protocol. | Vehicle mix and site geometry do not represent every local scene. | Split by sequence and report local-domain limitations. |
| Varaiya [10] | Max-pressure network signal control with theoretical properties under stated model assumptions. | Requires valid movement queues, downstream state, and lawful phases. | Use pressure as a transparent adaptive reference. |

| Study | Method and evidence | Limitations for Manzil | Design implication |
|---|---|---|---|
| Liang et al. [11] | Deep Q-learning for cycle control; demonstrated learned adaptation in a simulated setting. | Transfer across demand patterns and topology remains uncertain. | Bound actions and evaluate on held-out scenarios. |
| Wei et al. [12] | Reinforcement learning with pressure-grounded state and reward; exceeded tested baselines. | Findings depend on the reported networks, flows, and training setup. | Prefer a grounded reward to an arbitrary weighted sum. |
| Zheng et al. [13] | Phase-competition representation that learns relationships among competing movements. | Requires a more specialized phase encoding. | Preserve phase legality and structural invariance where feasible. |
| Wei et al. [14] | Graph-attention cooperation across signals; demonstrated coordination in tested simulations. | Training and attribution are more complex. | Keep the first network small and declare the coordination model. |
| Chu et al. [15] | Multi-agent deep reinforcement learning for larger networks and decentralized coordination. | Non-stationarity and tuning cost complicate reproducible comparison. | State ownership and communication explicitly. |
| Ma et al. [16] | High-throughput city-scale traffic-learning simulator intended for large experiments. | Migration would complicate alignment with the approved proposal. | Retain SUMO and document CityFlow as an alternative. |
| Lopez et al. [1] | Microscopic simulator architecture supporting detailed external control. | Network and car-following parameters limit external validity. | Version the network, demand, and simulator settings. |
| Dijkstra [17] | Nonnegative shortest-path search with a minimum-cost guarantee under its assumptions. | Represents a static snapshot unless costs are refreshed. | Use as the reference route implementation. |
| Hart et al. [3] | Heuristic minimum-cost search that guides exploration when the heuristic is valid. | An inadmissible heuristic can lose the optimality guarantee. | Use a free-flow-time lower bound or retain Dijkstra. |
| Humagain et al. [18] | Systematic review of emergency routing and preemption research. | A review does not validate any one implementation. | Measure emergency benefit and ordinary-traffic impact. |
| Qin and Khan [19] | Simulation of preemption entry and recovery strategies. | Reported parameters are not certified for Manzil's network. | Implement explicit safe-transition and recovery states. |
| Louati et al. [20] | Queue-aware multi-agent emergency crossing at signalized intersections. | Its multi-vehicle and multi-agent scope exceeds the core case. | Retain queue awareness and downstream checks. |

## 3.8 Conclusion

The review supports a conservative technical strategy in which evidence is built in stages. Transparent fixed-time and pressure-inspired controllers must operate correctly before a learned controller is compared with them, and measurement definitions must be fixed before any optimization result is interpreted. Video experiments should assess the complete observation chain rather than quote detector accuracy as a traffic-management result. Simulation experiments should use matched seeds, identical safety constraints, and held-out scenarios so that controller differences are not confused with different demand. Emergency evaluation must also separate the effect of route selection from the effect of signal priority, because combining both changes in one comparison would prevent a clear explanation of any improvement. The cited literature justifies the selected building blocks but does not establish their performance for Manzil's network, cameras, or computing resources. Chapter 4 therefore converts these findings into traceable requirements, while Chapter 5 defines the measurements and controlled comparisons needed to answer the research questions without claiming field deployment or certified safety evidence.

# Chapter 4 Software Requirement Specifications

This chapter translates the reviewed problem and evidence into a specification for the Manzil research prototype. The requirements describe observable behavior rather than implying that every item has already been implemented. Where the available repository evidence does not support a measured threshold, the text states a design target or requires profiling before a value is fixed. The specification gives particular attention to reproducibility, legal signal transitions, observation freshness, route-version consistency, and visible failure handling because these properties determine whether later experiments can be interpreted. It also separates the video-perception workflow from SUMO-based control experiments so that estimated observations are not silently presented as simulator ground truth. Account management, public dispatch integration, traffic enforcement, clinical response prediction, and physical signal deployment remain outside the approved scope. The resulting requirements provide a contract for the architecture in Chapter 6 and a traceable basis for the evaluation procedures in Chapter 5.

## 4.1 System Features

The prototype is organized into six feature groups that follow the operator's experimental workflow from configuration to evidence export. The grouping prevents a user-interface screen or implementation library from being mistaken for a requirement and clarifies which service owns each outcome. It also shows that emergency operation is not an isolated routing feature: a valid mission depends on scenario state, traffic observations, signal control, monitoring, and durable records. The following statements summarize those groups before the individual functional requirements define testable behavior.

1. The operator can select a network, demand profile, controller, seed, incident, duration, and optional emergency mission before a run begins.
2. The perception workflow can process licensed recorded video into detections, tracks, road-region observations, and timestamped aggregate measures with confidence information.
3. The simulation workflow advances SUMO through one owner, reads traffic state, applies fixed or adaptive requests, and preserves legal phase transitions.
4. The emergency workflow validates endpoints, computes a dynamic path, applies stable rerouting, activates bounded moving priority, detects passage, and recovers normal control.
5. The monitoring interface presents map state, signals, traffic indicators, ambulance route, corridor state, warnings, and the current data-source mode.
6. The experiment service retains immutable configuration, events, metrics, software versions, seeds,

and termination state for comparison and export.

## 4.2 Functional Requirements

Table 4.1 defines the mandatory system behavior using stable identifiers that are reused in the traceability matrix. The requirements distinguish a requested signal action from the action actually applied, require route and observation versions to be retained, and make fallback behavior part of the specification rather than an informal implementation detail. Several entries use the word "shall" because they are acceptance conditions for the proposed prototype; they are not claims of completed implementation. Together, FR-01 through FR-08 cover experiment configuration and normal control, FR-09 through FR-14 define the emergency-route and corridor lifecycle, and FR-15 through FR-20 cover monitoring, evidence, comparison, export, and controller failure. This arrangement lets tests target a coherent workflow while preserving individual identifiers for defect tracking. A requirement is satisfied only when its normal and failure paths are demonstrated with retained evidence. For example, returning a path is insufficient for FR-10 if the cost snapshot and graph-validity conditions are not also recorded.

**Table 4.1: Functional requirements**

| ID | Requirement |
|---|---|
| FR-01 | The system shall validate a scenario configuration before starting and shall reject missing network, controller, seed, duration, or demand inputs. |
| FR-02 | The system shall start, pause, resume, and stop one controlled SUMO run while preserving an unambiguous run state. |
| FR-03 | The simulation adapter shall be the sole owner of SUMO stepping and signal actuation. |
| FR-04 | The perception worker shall process a selected recorded video and emit timestamped detections, tracks, source metadata, and processing status. |
| FR-05 | The observation service shall aggregate valid vehicle observations into defined road, lane, or approach intervals without silently mixing sources. |
| FR-06 | The operator shall be able to select fixed-time, rule-adaptive, or the configured RL controller before a run. |
| FR-07 | Every signal action shall identify its intersection, requested and applied phase, effective simulation step, controller, and reason. |
| FR-08 | The system shall enforce minimum green and configured yellow/all-red clearance before incompatible movements receive right of way. |

| ID | Requirement |
|---|---|
| FR-09 | The emergency service shall validate ambulance origin and destination and return a clear no-route response if either is invalid or disconnected. |
| FR-10 | The route service shall compute a valid remaining path using nonnegative time-based edge costs and shall record the cost snapshot. |
| FR-11 | The rerouting service shall re-evaluate the remaining path after a closure, invalidated edge, or sustained cost improvement above the configured threshold. |
| FR-12 | The corridor coordinator shall activate priority only for a bounded look-ahead set of intersections on the current ambulance route. |
| FR-13 | The corridor coordinator shall detect passage or cancellation and restore the affected intersection to a safe normal-control state. |
| FR-14 | A route revision shall cancel priority requests that no longer belong to the remaining path before activating new ones. |
| FR-15 | The dashboard shall display the current source mode, simulation status, controller, signal state, traffic indicators, ambulance position, route, and corridor state. |
| FR-16 | The dashboard shall present stale, invalid, disconnected, and fallback states visibly rather than continuing with apparently current data. |
| FR-17 | The system shall record configuration, seed, component versions, periodic metrics, route revisions, signal events, failures, and termination reason for each run. |
| FR-18 | The comparison service shall align selected runs by metric definition and warn when network, demand, duration, warm-up, or measurement versions are incompatible. |
| FR-19 | The operator shall be able to export run configuration and metrics in a documented machine-readable format. |
| FR-20 | On loss of the learned controller or invalid action, the control service shall fall back to a configured safe baseline and record the transition. |

## 4.3 Quality Attributes and Non-Functional Requirements

Functional behavior alone would permit a demonstration that appears successful but cannot be reproduced or audited. Table 4.2 therefore states the quality attributes that constrain every feature group. Reproducibility and data integrity require a run to preserve the exact scenario, version, seed, and metric definition used to generate a result. Safety requirements apply within simulation and prevent controllers

from bypassing clearance or ownership rules. Freshness and isolation ensure that a stale perception result or slow dashboard cannot silently influence a later simulation step. Responsiveness is intentionally expressed as a profiled design target because the repository contains no defensible hardware benchmark from which to invent a deadline. Privacy is limited to the handling of licensed or team-authorized video and does not imply a production governance framework. These non-functional requirements also define how negative outcomes are handled: an unavailable controller, disconnected simulator, or incomplete record must remain visible rather than being removed from the evaluation set without explanation.

**Table 4.2: Non-functional requirements and design targets**

| ID | Attribute | Requirement or target |
|---|---|---|
| NFR-01 | Reproducibility | A run shall store network and demand identifiers, seed, controller configuration, software/model versions, warm-up, duration, and metric schema. |
| NFR-02 | Safety in simulation | Incompatible green movements shall never be commanded intentionally; all emergency transitions shall use the configured clearance sequence. |
| NFR-03 | Auditability | Applied decisions shall be distinguishable from requests, and their simulation timestamps and reasons shall be retained. |
| NFR-04 | Freshness | Each observation shall carry produced-at and effective simulation time. Stale thresholds are configuration values and stale data shall not silently drive control. |
| NFR-05 | Responsiveness | The control action for step $t$ should be available before its configured decision deadline. The exact deadline is a design target to be set after baseline profiling, not an achieved value. |
| NFR-06 | Isolation | Background video inference and dashboard clients shall not advance SUMO or directly actuate signals. |
| NFR-07 | Usability | A trained operator should be able to configure the core demonstration from one experiment screen, with units and validation messages visible. |
| NFR-08 | Portability | Services should be runnable from versioned environments; containerization is proposed after local baseline integration is stable. |

| ID | Attribute | Requirement or target |
|---|---|---|
| NFR-09 | Maintainability | Component interfaces shall use typed, versioned data contracts and controller implementations shall share one interface. |
| NFR-10 | Data integrity | A completed run configuration shall be immutable; derived summaries shall retain their metric version and source run identifiers. |
| NFR-11 | Privacy | Video shall be limited to licensed or team-authorized material, stored only as needed, and excluded from exports unless explicitly requested. |
| NFR-12 | Availability | The prototype shall fail visibly and recover to a safe stopped or baseline-controlled simulation state; no public-service availability is claimed. |

## 4.4 Assumptions and Dependencies

SUMO and TraCI are required for controlled traffic experiments [1, 2]. A selected network must contain legal lane connections and signal plans after import and manual cleanup, and representative routes must be validated before it is used for comparison. Video sources require sufficient visibility, a documented license or team authorization, and annotations appropriate to the selected measure. The simulation assumes one clock owner, nonnegative route costs, and emergency endpoints on ambulance-accessible edges. It also assumes that configured phase transitions reflect the validated simulation network; they are not certified timings for physical signal hardware. The dashboard is an operator and evaluator tool rather than an autonomous public dispatch system, and a displayed route does not constitute operational navigation advice. When an assumption cannot be met, the affected workflow must stop or disclose a bounded fallback. It must not silently substitute a different network, data source, time unit, or observation mode because that would make later comparisons irreproducible.

React and FastAPI are fixed by the approved proposal for the interface and application service boundary [21, 22]. Python is implied by the planned perception, reinforcement-learning, and TraCI ecosystem. PostgreSQL, a WebSocket transport, Docker, and specific chart or map libraries are proposed dependencies and remain replaceable until implementation confirms them. These assumptions are deliberately narrower than a production deployment specification. Manzil is evaluated as a controlled prototype on prepared networks, not as an interface to public signal controllers or ambulance dispatch infrastruc-

ture. A selected video source must have suitable permission and sufficient visibility for the intended annotation task, while the simulation network must be checked for lane connectivity, valid traffic-light programs, and ambulance-accessible routes. If any assumption fails, the run must be rejected or marked with an explicit limitation instead of substituting an undocumented configuration. This treatment keeps dependencies traceable and prevents a convenient library choice from becoming an unsupported architectural claim.

## 4.5 Hardware and Software Requirements

Table 4.3 separates development and training needs from the smaller set of resources required for a repeatable demonstration. This distinction matters because detector training or reinforcement-learning experiments may benefit from a CUDA-capable GPU, while a prepared SUMO scenario and dashboard may run acceptably on a CPU after models and policies have been produced. Exact memory, storage, and timing values remain unconfirmed until the team profiles the selected network, video, and model; the table therefore states capabilities rather than fabricated minimum specifications. The demonstration environment must use the same versioned configuration and metric schema as the evaluation runs, even when it uses fewer scenarios. A file-backed store is acceptable during early integration, but migration to PostgreSQL remains proposed when cross-run integrity and concurrent access justify it. In either case, the report treats the schema and provenance rules as requirements independent of a particular database engine. Measured resource use will replace these capability statements only after the selected workload is stable.

**Table 4.3: Development and demonstration environment**

| Area | Development/training need | Demonstration need |
|---|---|---|
| Compute | Multi-core CPU; CUDA-capable GPU is desirable for detector/RL experiments but exact memory is unconfirmed | CPU able to run selected SUMO network and services; GPU only if live video inference is demonstrated |
| Memory/storage | Sufficient for environments, video, checkpoints, run logs, and repeated traces; capacity to be measured | Selected scenario, service builds, model, and demonstration records |
| Core software | Python, SUMO/TraCI, detector/tracker framework, RL library, Node.js/React toolchain, Git | Version-pinned runtime and browser |
| Data tools | OpenStreetMap import, SUMO NetEdit, annotation tooling | Prepared network and approved demonstration clips |
| Persistence | File-backed prototype acceptable initially; PostgreSQL proposed for structured records | Same schema/version as evaluation records |

## 4.6 Use Cases

The primary actor is a research operator who configures experiments, observes their progress, initiates a simulated emergency mission, and exports evidence. SUMO, recorded video, and the experiment store support those actions but do not independently choose research parameters. Evaluators consume dashboard views and exported records without receiving control privileges. Figure 4.1 presents this boundary and connects the operator to eight goal-oriented interactions. Configuration and simulation appear separately because validation must finish before the single clock owner starts a run. Dispatch, corridor monitoring, and rerouting are also distinct so that a route revision cannot be treated as an invisible side effect. Tables 4.4–4.11 then specify preconditions, triggers, normal flows, failures, and postconditions for each interaction. Collectively, they test more than the successful interface path: invalid endpoints, stale streams, no-route cases, controller failure, incompatible comparisons, and termination reasons must all produce observable outcomes. The use cases therefore define acceptance evidence as well as operator

interaction.

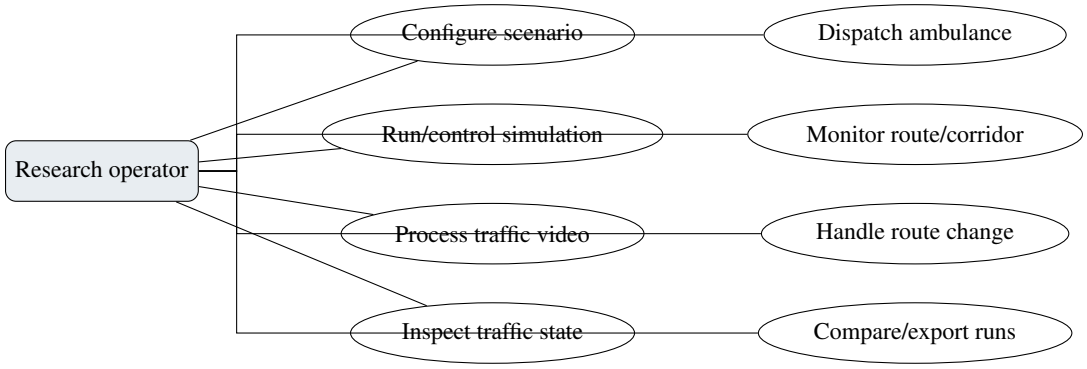



**Figure 4.1: Proposed operator use-case diagram**

**Table 4.4: UC-01 Configure a scenario**

| | |
|---|---|
| **Name / actor** | UC-01 Configure a scenario / Research operator |
| **Preconditions** | Network and controller descriptors are available. |
| **Trigger** | Operator requests a new experiment. |
| **Main success flow** | Select network, demand, controller, seed, warm-up, duration, source mode, incident, and optional mission; validate; store a draft configuration. |
| **Alternative/failure flow** | Reject incompatible controller/network, invalid duration, missing seed, or inaccessible input with field-specific messages. |
| **Postconditions** | A valid, versioned configuration is ready; no run has started. |

**Table 4.5: UC-02 Run and control a simulation**

| | |
|---|---|
| **Name / actor** | UC-02 Run and control a simulation / Research operator |
| **Preconditions** | UC-01 is valid; SUMO adapter is available. |
| **Trigger** | Operator selects Start. |
| **Main success flow** | Create run record; launch SUMO; complete warm-up; step the simulation; collect state; request and apply lawful actions; publish snapshots; stop at terminal condition. |
| **Alternative/failure flow** | If launch/TraCI fails, mark failed and retain logs. Pause/resume does not create competing clock owners. Invalid controller action invokes fallback. |
| **Postconditions** | Run is completed, stopped, or failed with an explicit reason. |

**Table 4.6: UC-03 Process traffic video**

| | |
|---|---|
| **Name / actor** | UC-03 Process traffic video / Research operator |
| **Preconditions** | Licensed video and model configuration are available. |
| **Trigger** | Operator selects a video job. |
| **Main success flow** | Validate source; decode frames; detect vehicles; associate tracks; map valid tracks to regions; aggregate interval observations; store metrics and job status. |
| **Alternative/failure flow** | Unreadable media, unsupported format, model failure, missing calibration, or low-confidence observation is reported; physical queue length is omitted when calibration is unavailable. |
| **Postconditions** | Completed observations identify source, model, interval, units, and confidence. |

**Table 4.7: UC-04 Inspect traffic state**

| | |
|---|---|
| **Name / actor** | UC-04 Inspect traffic state / Research operator |
| **Preconditions** | A run or perception job has emitted data. |
| **Trigger** | Operator opens overview or intersection detail. |
| **Main success flow** | Display source mode, freshness, counts, queues/proxies, speeds, phases, controller, and warnings; allow time/intersection selection. |
| **Alternative/failure flow** | Disconnected or stale streams remain visibly marked; last value is not presented as current. |
| **Postconditions** | No system decision is changed; the inspected state is traceable to its source. |

**Table 4.8: UC-05 Dispatch an ambulance**

| | |
|---|---|
| **Name / actor** | UC-05 Dispatch an ambulance / Research operator |
| **Preconditions** | A running emergency-enabled scenario exists. |
| **Trigger** | Operator submits origin and destination. |
| **Main success flow** | Validate accessible edges; create mission; compute route/cost; assign simulated ambulance; publish route; request bounded corridor look-ahead. |
| **Alternative/failure flow** | Invalid edge or no route returns a reason without partial priority. Duplicate active mission is rejected in the core single-ambulance mode. |
| **Postconditions** | Mission is active with a valid current route, or rejected cleanly. |

**Table 4.9: UC-06 Monitor route and corridor**

| | |
|---|---|
| **Name / actor** | UC-06 Monitor route and corridor / Research operator |
| **Preconditions** | An emergency mission is active. |
| **Trigger** | Ambulance advances through the network. |
| **Main success flow** | Update position and remaining path; request safe transition ahead; display corridor states; detect passage; release and recover each intersection. |
| **Alternative/failure flow** | Missing vehicle, stale position, unsafe request, or corridor timeout triggers cancellation/fallback and a visible alert. |
| **Postconditions** | Mission progress and all priority transitions are recorded. |

**Table 4.10: UC-07 Handle blockage or changed route**

| | |
|---|---|
| **Name / actor** | UC-07 Handle blockage or changed route / Research operator |
| **Preconditions** | An active mission has a feasible current position. |
| **Trigger** | Closure or sustained cost change triggers evaluation. |
| **Main success flow** | Recompute path from a valid remaining edge; compare benefits; if threshold is met, atomically revise route, cancel obsolete requests, and activate new look-ahead. |
| **Alternative/failure flow** | If improvement is too small, retain route. If no route exists, cancel upcoming priority and mark mission blocked. |
| **Postconditions** | Exactly one route version is current; every revision has cause and cost snapshot. |

**Table 4.11: UC-08 Compare and export runs**

| | |
|---|---|
| **Name / actor** | UC-08 Compare and export runs / Research operator |
| **Preconditions** | At least two completed or failed run records exist. |
| **Trigger** | Operator selects runs and metrics. |
| **Main success flow** | Check comparability; align metric definitions; present distributions and configuration differences; export selected records. |
| **Alternative/failure flow** | Warn or refuse direct comparison when network, demand, duration, warm-up, or metric version differs. Failed runs are not silently omitted. |
| **Postconditions** | Comparison and export identify source run IDs and schema version. |

## 4.7 Proposed Graphical User Interface

The following figures are proposed wireframes rather than screenshots of implemented software. They validate information hierarchy, source disclosure, error visibility, and the actions needed for a controlled demonstration without fabricating implementation progress. Figure 4.2 gives the operator a network-level view in which the selected run, controller, source mode, ambulance route, corridor lifecycle, and warnings remain visible together. This arrangement helps prevent a plausible map animation from hiding stale data or a fallback controller. Figure 4.3 narrows the view to one intersection and pairs a processed-video preview with model provenance, freshness, regional measures, confidence, and invalid-state indicators. The calibration field is explicit because pixel-derived queue proxies must not be mislabeled as physical queue length. Both designs use color only as a supporting cue; labels retain the meaning required for accessibility and for printed evidence. The reader can consequently distinguish authoritative values, estimates, and warnings without relying on animation.

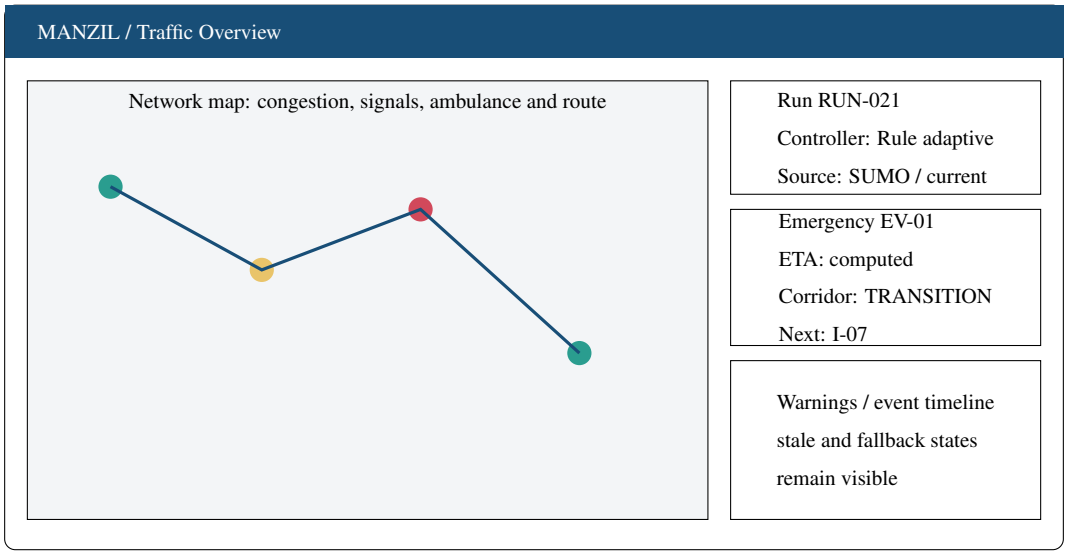



**Figure 4.2: Proposed traffic overview wireframe**

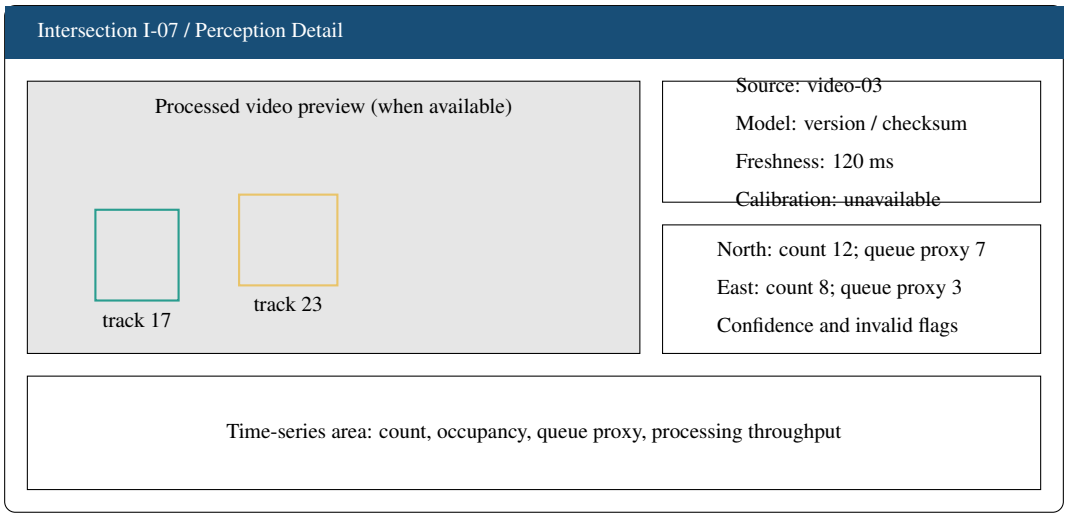



**Figure 4.3: Proposed perception and intersection-detail wireframe**

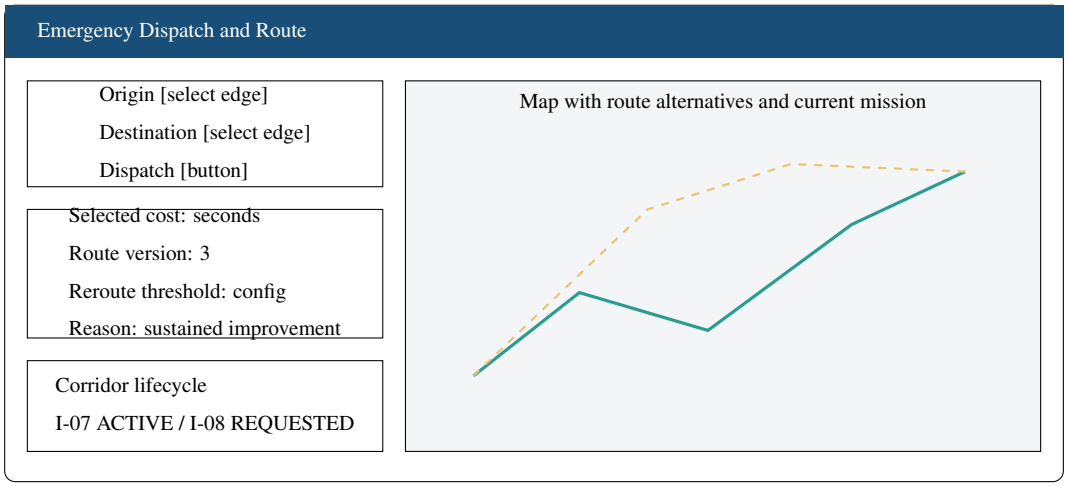



**Figure 4.4: Proposed emergency dispatch and route wireframe**

Emergency operation and experiment setup require different decisions, so they are separated in Figures 4.4 and 4.5. The dispatch view shows validated endpoints, the current route version, its time-based cost, the reason for a revision, and the state of intersections in the bounded look-ahead corridor. It also permits a candidate route to be shown without implying that every alternative has been activated. The configuration view collects the network, demand, controller, seed, duration, incident, mission, and metric schema before a run begins. Its validation panel exposes unresolved identifiers, illegal phase plans, absent calibration, or checkpoint mismatch, and the start action remains unavailable until mandatory errors are resolved. The two wireframes therefore support auditability at different stages: one explains a live mission decision, while the other preserves the conditions under which that decision can later be compared. Neither view grants direct signal control, bypasses backend validation, or changes an active run's immutable configuration.

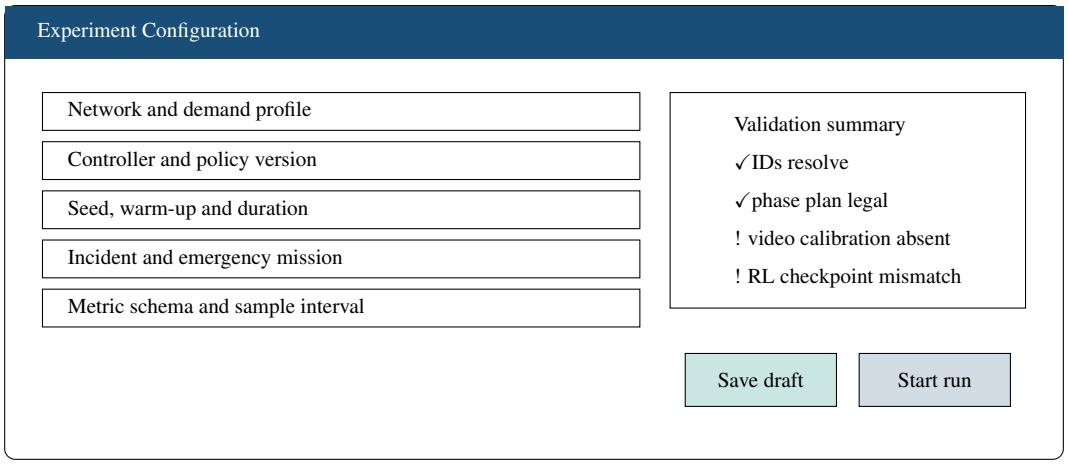



**Figure 4.5: Proposed experiment-configuration wireframe**

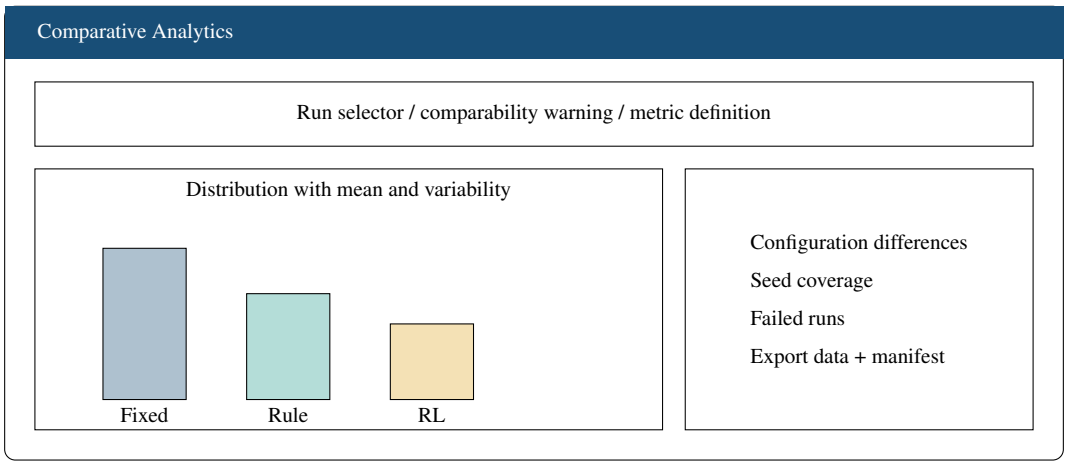



**Figure 4.6: Proposed comparative-analytics wireframe**

Figure 4.6 represents the final evidence-review workflow. Run selection is paired with a comparability warning and metric definition so that two visually adjacent bars are not assumed to represent matched experiments. The chart area is schematic and contains no claimed result; an implemented view would present the chosen distribution, central tendency, variability, and sample count from retained run records. The adjoining panel exposes configuration differences, seed coverage, failed runs, and export controls. This design responds directly to the methodology requirement that failed or incompatible runs must not disappear from a favorable summary. It also keeps measurement units and metric versions close to the visualization, reducing the risk that a screenshot is detached from its experimental meaning. Together, the five wireframes define the minimum information architecture for the prototype while leaving visual polish and specific chart libraries open to implementation. Exported graphics must retain the same metric and run identifiers.

## 4.8 Data Design

Persistent data is reserved for configuration, observations, decisions, missions, and outcomes that require audit or comparison. The live road graph, current TraCI subscription, and frame buffers remain in-memory runtime state because they can be reconstructed or are too transient to justify row-level storage. Raw video remains an external file identified by provenance and checksum rather than a database blob. Figure 4.7 presents the proposed record relationships: a scenario produces multiple runs; a run owns observations, decisions, metrics, and at most one active core mission; and each mission can produce several immutable route versions whose priority events retain the version that justified them. This version link is essential when a reroute cancels obsolete requests. Table 4.12 complements the diagram with key fields and integrity rules, including immutable run configuration, mandatory units and source mode, monotonically increasing route versions, and matching metric versions for comparison. The model describes proposed persistence responsibilities and does not claim that a production database has already been populated.

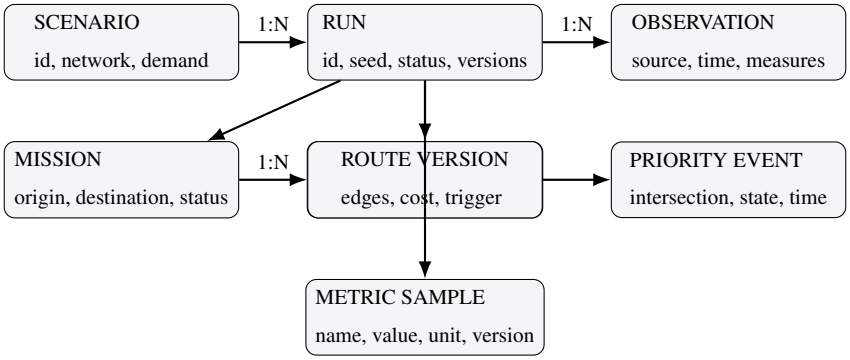



**Figure 4.7: Proposed experiment-record entity relationship model**

**Table 4.12: Data dictionary**

| Entity | Key fields | Integrity rule |
|---|---|---|
| Scenario | scenario_id, network_version, demand_version | References immutable network and demand manifests. |
| Run | run_id, scenario_id, seed, controller_version, status | Configuration becomes immutable after start; terminal reason required. |

| Entity | Key fields | Integrity rule |
|---|---|---|
| Observation | run_id/job_id, source_mode, entity_id, observed_at, valid_until, values, units | Source and units are mandatory; validity interval cannot be inferred. |
| ControlDecision | run_id, intersection_id, requested_phase, applied_phase, effective_step, reason | Requested and applied actions are separate fields. |
| Mission | mission_id, run_id, vehicle_id, origin, destination, status | Only one active core mission per run. |
| RouteVersion | mission_id, version, edge_list, cost_seconds, trigger, created_at | Version increases monotonically; path begins at valid current position. |
| PriorityEvent | mission_id, route_version, intersection_id, state, requested/applied time | References the route version that justified priority. |
| MetricSample | run_id, metric_name, metric_version, value, unit, interval | Comparisons require matching metric name, version, and unit. |
| Artifact | run/job_id, type, URI, checksum, license/provenance | Content is addressable and provenance is retained. |

## 4.9 Risk Analysis

Table 4.13 records technical and research risks that could invalidate the prototype or its conclusions. Likelihood and impact are qualitative planning judgments rather than measured probabilities. The highest-consequence risks concern invalid signal phases, irreproducible results, unsuitable map imports,

and evidence that cannot be legally or reliably used. Each mitigation is paired with a bounded fallback so that the team can preserve an honest deliverable if a preferred component fails. For example, a documented synthetic network is preferable to an unverified real-map extract, and a negative reinforcement-learning result is preferable to tuning on the final test scenarios. The register also treats timing drift, route oscillation, occlusion, and reward imbalance as system-level concerns rather than isolated coding defects. These risks are revisited through invariant tests, scenario checks, provenance records, and the controlled comparisons defined in Chapter 5; unresolved items must remain visible in the final limitations rather than being converted into unsupported performance claims.

**Table 4.13: Risk register**

| Risk | Likelihood | Impact | Mitigation | Fallback |
|---|---|---|---|---|
| Map import lacks valid signals/turns | Medium | High | Inspect and clean a small extract early; validate routes and phases | Use a documented synthetic grid |
| Local video lacks labels or permission | Medium | High | Use licensed public sequences and record provenance | Evaluate public data only; state domain gap |
| Occlusion causes unstable counts | High | Medium | Region rules, temporal aggregation, tracker comparison | Limit claims to labelled conditions |
| RL fails to converge or generalize | Medium | High | Stable baselines, bounded action space, tune only on training scenarios | Report baseline comparison; treat RL as negative/partial finding |
| Reward improves one metric unfairly | Medium | High | Multi-metric evaluation and starvation constraints | Reject policy violating safety or fairness checks |
| Rerouting oscillates | Medium | Medium | Benefit threshold, persistence, cooldown, closure exception | Freeze current feasible route |
| Preemption leaves unsafe phase | Low | Very high | Central state machine and invariant tests | Cancel request and use safe baseline transition |

| | | | | |
|---|---|---|---|---|
| Service timing drifts from simulation | Medium | High | One step owner, simulation timestamps, bounded queues | Run synchronously without live perception |
| Insufficient compute | Medium | Medium | Profile early; smaller network/model and offline training | Demonstrate recorded inference and compact scenarios |
| Results are not reproducible | Medium | High | Immutable manifests, seeds, versioned metrics and exports | Exclude untraceable runs |

## 4.10 Requirements Traceability

The traceability matrix in Table 4.14 connects requirement groups to the use cases, design modules, and planned verification evidence that will demonstrate them. Grouping closely related identifiers keeps the table readable while preserving the detailed statements in Tables 4.1 and 4.2. The matrix reveals several cross-cutting obligations. Reproducibility depends on the experiment service and store rather than on the dashboard alone; legal control depends on both the orchestrator and safety filter; and emergency correctness spans routing, corridor lifecycle, simulation actuation, and event persistence. Verification consequently combines schema checks, unit and invariant tests, small-network integration tests, failure injection, annotated video evaluation, and matched experimental runs. A successful user-interface walkthrough cannot replace these checks. Conversely, an algorithmic metric cannot satisfy usability or failure-visibility requirements. The matrix will be updated with test identifiers during implementation so that incomplete or failed evidence remains traceable to the affected requirement. This update will not alter the stable requirement identifiers.

**Table 4.14: Requirements traceability matrix**

| **Requirements** | **Use cases** | **Design modules** | **Planned verification** |
|---|---|---|---|
| FR-01, 17–19; NFR-01,10 | UC-01, UC-08 | Experiment service, store | Schema, validation, replay and comparison tests |

| Requirements | Use cases | Design modules | Planned verification |
|---|---|---|---|
| FR-02,03,07; NFR-02,06 | UC-02 | Simulation adapter, control orchestrator | Single-owner and phase-invariant tests |
| FR-04,05; NFR-04,11 | UC-03, UC-04 | Perception worker, state aggregator | Annotated sequence evaluation and stale-data tests |
| FR-06,08,20 | UC-02 | Controller interface, safety filter | Fixed scenarios, illegal-action and fallback tests |
| FR-09–11,14 | UC-05, UC-07 | Graph/routing service | Disconnected graph, closure, hysteresis and path-validity tests |
| FR-12,13 | UC-06, UC-07 | Corridor coordinator | Lifecycle, timeout, cancellation and recovery tests |
| FR-15,16 | UC-04, UC-06 | API gateway, dashboard | UI state and warning acceptance checks |
| NFR-03,05,07–09,12 | All | Cross-cutting contracts and deployment | Log audit, profiling, usability walkthrough, failure injection |

## 4.11 Conclusion

The specification centers on traceability and failure visibility rather than on a collection of loosely connected features. It requires one simulation-clock owner, legal signal transitions, stable route revision, explicit observation freshness, immutable run configuration, and comparisons that cannot hide incompatible settings or failed trials. Functional requirements define what the operator and services must observe, while non-functional requirements constrain timing, ownership, privacy, auditability, and recovery. The use cases exercise both successful and failed paths, the wireframes show how critical state and provenance remain visible, and the data model preserves the route and decision history needed for later analysis. Risks and traceability link those requirements to bounded fallbacks and planned verification. Chapter 5 now converts the specification into a staged research protocol, and Chapter 6 assigns each mutable resource and interface to a clear architectural owner. Implementation status will continue

to be reported separately so that a specified feature is never presented as completed evidence.

# Chapter 5 Proposed Approach and Methodology

This chapter defines how Manzil will be constructed and evaluated so that the research questions can be answered with controlled evidence rather than a favorable demonstration run. The method begins by freezing definitions, exclusions, baselines, and data partitions before the final comparison. It then validates the network, observations, controllers, routing, and corridor lifecycle in stages, because a result from an integrated run is not interpretable when an underlying measurement or safety transition is still uncertain. Figure 5.1 summarizes this sequence from preparation to reporting. Each stage produces an artifact or acceptance check that is retained with the experiment manifest. The protocol also separates detector evaluation from signal-control evaluation: recorded video supplies annotated evidence for perception, while SUMO supplies controlled vehicle and signal state for the controller, route, and emergency experiments. This separation prevents estimated video quantities from being passed off as exact network state and allows an optional observation-noise study only after both baselines are stable.

## 5.1 Overall Research Workflow

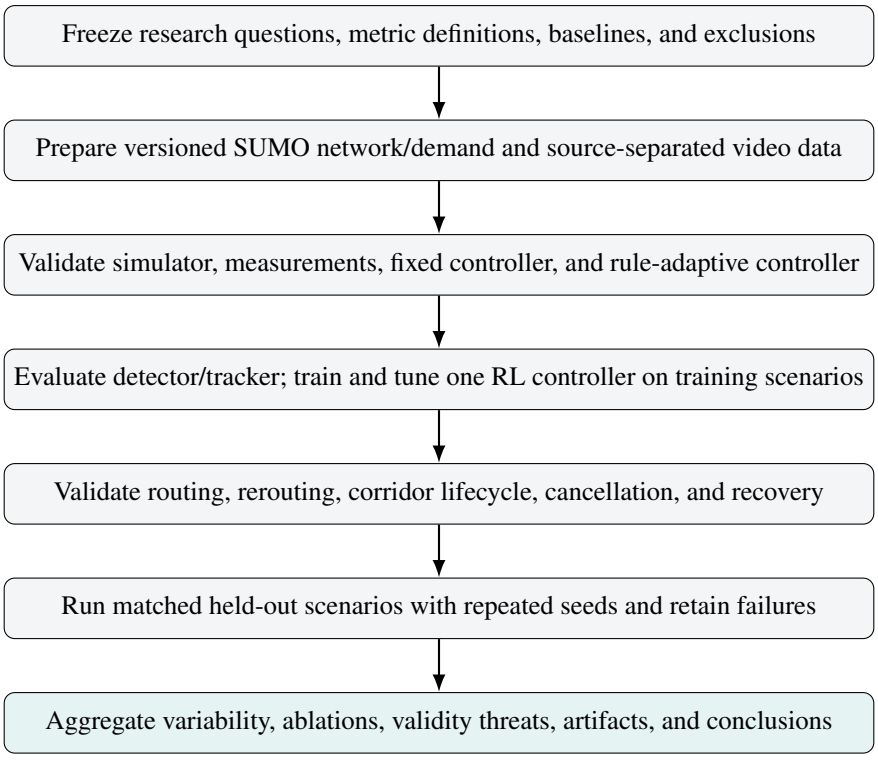



**Figure 5.1: Proposed research and evaluation workflow**

The workflow in Figure 5.1 is ordered to prevent later components from masking earlier faults. Research questions and metric definitions are fixed before data preparation; the network and demand are validated before controllers are compared; and ordinary traffic control is established before emergency routing and

priority are enabled. Two input modes are retained throughout this process. In video mode, annotations provide the reference for perception metrics, and no claim is made that a selected clip determines the full state of the simulated network. In simulation mode, SUMO supplies lane, vehicle, signal, and route state for controlled experiments. Both modes use the same typed traffic-state concepts but retain a source field, units, validity interval, and confidence where applicable. A later noise experiment may inject measured perception error into simulation observations, but it will be included only after the baseline protocol is complete and its parameters are fixed in advance. The final stage reports variability, failures, ablations, and validity limits alongside successful runs.

## 5.2 Network and Scenario Preparation

### 5.2.1 Network Provenance and Validation

A selected, manageable Lahore road area will be exported from OpenStreetMap under its applicable attribution and license conditions [23]. The exact bounding box, extraction date, source file checksum, and conversion command will be stored so that a later change in the public map cannot silently alter an experiment. SUMO's import toolchain will convert the graph, after which NetEdit inspection will verify lane counts, legal turns, junction types, traffic-light programs, pedestrian-only edges, disconnected components, and ambulance-accessible routes. Automated checks will then attempt representative routes, enumerate signal phases, and reject movements that conflict with the intended vehicle classes. The initial planning range is approximately 10–20 signalized intersections, but this is not a confirmed network fact or a requirement to retain a defective import. A smaller validated extract is preferable to a larger inaccurate one because route continuity and lawful signal control are prerequisites for every later result. If the chosen map cannot be stabilized within the schedule, a documented synthetic multi-intersection grid becomes the fallback, and all results will be identified as synthetic rather than generalized to Lahore roads.

### 5.2.2 Demand and Incidents

Demand files will define light, medium, and heavy conditions together with an uneven-approach case that can expose starvation. Vehicle insertion uses explicit random seeds, and a declared warm-up interval allows traffic to populate the network before measurement begins. Incident scenarios close an edge, reduce lane capacity, or introduce a controlled obstruction at a fixed simulation time. A failed insertion caused by a full lane is retained as an outcome rather than retried until the run appears favorable. The same requested demand and seed are replayed across controllers where SUMO semantics permit. When controller behavior changes route choice or insertion feasibility, the manifest distinguishes the

original demand request from actual departures so that throughput is not interpreted without its denominator. Scenario families are separated into development, tuning, and held-out evaluation sets before final runs. Origin-destination pairs, incident timing, and emergency dispatch time are also matched within each comparison. These controls make differences attributable to the selected policy while preserving congestion effects that legitimately arise from that policy.

## 5.3 Video Perception Pipeline

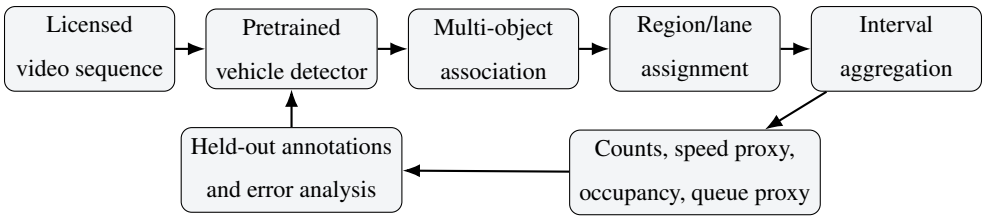



**Figure 5.2: Proposed traffic-video perception pipeline**

Figure 5.2 presents the complete video-observation chain rather than depicting detection as an isolated result. The detector produces a class, confidence, and image-space box; the tracker associates observations through time; and a bottom-center or calibrated road-contact point maps each valid track to a polygonal lane or approach region. Directed line-crossing or region-entry rules use track history to avoid counting the same vehicle in every frame. Interval aggregation then derives counts, image occupancy, a queue proxy, and a speed proxy with source time and confidence. Held-out annotations feed the error-analysis path and are never used to tune the final evaluation sequence. A speed in physical units requires camera calibration or a defensible known scale. Without that evidence, the system reports pixel displacement or a categorical slow or stationary state rather than kilometres per hour. The figure therefore emphasizes where uncertainty enters the pipeline and why an apparently stable box is not yet a validated traffic measure.

For a lane region $l$ over interval $k$, a simple occupancy proxy is

$$O_{l,k} = \frac{1}{T_k} \sum_{t \in k} \frac{\sum_{i \in V_{l,t}} A(B_{i,t} \cap R_l)}{A(R_l)}, \tag{5.1}$$

where $B_{i,t}$ is a tracked box, $R_l$ is the image region, and $T_k$ is the number of sampled frames. This is image occupancy, not road density. A queue proxy counts stable tracks inside a defined upstream region whose image-motion or calibrated speed is below a threshold for a minimum persistence time.

The detector will be initialized from a documented pretrained checkpoint, consistent with the practical motivation of YOLO-family detection [5]. ByteTrack is the leading tracker candidate because its associ-

ation of lower-confidence boxes may reduce fragmentation under temporary occlusion [7]. DeepSORT remains a comparison when appearance information materially improves association in the selected scenes [6]. Thresholds are tuned only on development sequences and then frozen for held-out evaluation. Selection considers interval count error, track fragmentation when identity is required, queue-proxy error, processing throughput, and failure by scene condition; benchmark reputation alone is insufficient. Equation (5.1) defines image occupancy when calibration is unavailable, using the fraction of a declared region covered by tracked boxes across sampled frames. Because perspective and occlusion affect that fraction, it is reported as a proxy with its region geometry and sampling interval, not renamed as physical road density. This evaluation keeps the claimed output aligned with what the video can actually support.

## 5.4 Normal Traffic Controllers

### 5.4.1 Fixed-Time Controller

The fixed controller will use a declared cycle and phase sequence appropriate to each selected junction. Green, yellow, all-red, cycle, and offset values will be versioned with the scenario, and the same movement definitions and clearance rules will constrain every competing controller. The baseline will not be intentionally starved or assigned unrealistic timing to make adaptation appear superior. Initial green splits will be selected from the declared demand and checked against approximate approach service so that at least the light scenario is feasible and heavier scenarios create interpretable congestion rather than immediate gridlock. Its deterministic schedule makes it useful for validating demand generation, metric collection, ambulance route ablations, and replay before learning is introduced. If a run violates phase invariants or produces implausible counts under fixed control, adaptive and reinforcement-learning comparisons are postponed until the cause is resolved. The baseline therefore serves as both a comparator and a diagnostic reference, not as a weak opponent constructed for a favorable result.

### 5.4.2 Rule-Based Adaptive Controller

The initial transparent controller evaluates only legal candidate phases. A pressure-inspired score for phase $p$ is

$$S_p(t) = \sum_{(l,m)\in p} (q_l(t) - \alpha q_m(t)) + \beta w_l(t), \tag{5.2}$$

where $q_l$ and $q_m$ are upstream and downstream queues for a served movement, $w_l$ is the maximum accumulated wait among its approaches, and $\alpha, \beta$ are documented configuration values. This follows the motivation of max-pressure control [10] while retaining a direct anti-starvation term.

Equation 5.2 assigns a transparent score to each legal candidate phase using upstream demand, down-

stream congestion, and the longest accumulated approach wait. The downstream term discourages serving a movement whose receiving link lacks space, while the wait term provides a direct safeguard against repeatedly postponing a low-volume approach. Coefficients are configuration values established on development scenarios and retained with the run manifest. At each decision instant, the current green is held until minimum green has elapsed. It may then be extended when it remains the preferred phase and maximum green has not been reached. A phase whose maximum-red limit is approached receives explicit promotion, and every switch inserts the same yellow and all-red sequence used by other controllers. The controller returns a requested phase and its contributing values; it does not actuate SUMO directly. This explanation record permits a reviewer to distinguish an algorithmic request from a safety-filter transformation and to diagnose whether observed performance came from pressure, fairness promotion, or repeated intervention.

### 5.4.3 Proposed RL Formulation

One reinforcement-learning formulation will be selected only after the fixed and rule-based baselines are stable. A feasible initial choice is a discrete-action value-based policy for a small, fixed phase set; pressure-informed reward design is supported by PressLight [12]. The action space requests one legal phase or a bounded extension, while the safety layer retains authority over minimum green and clearance. If the chosen network requires coordination and the training budget permits it, centralized observations with per-intersection actions are preferable to loosely defined independent learners because ownership and shared information remain explicit. Model architecture, observation normalization, replay settings, exploration schedule, optimizer, training budget, and checkpoint-selection rule will be recorded before held-out evaluation. This design does not assume that reinforcement learning will outperform the transparent baseline. If it fails to converge, violates fairness constraints, or offers no consistent held-out benefit, that outcome remains part of the research evidence and the working rule-based controller remains the operational fallback.

The proposed state for each decision interval includes normalized incoming and outgoing queues, approach waiting time, occupancy, current phase, elapsed phase time, and an emergency-mode flag. An action requests a legal phase or a bounded extension. The safety layer remains authoritative. A candidate reward is

$$r_t = -\lambda_p P_t - \lambda_w W_t - \lambda_s X_t - \lambda_c C_t, \tag{5.3}$$

where $P_t$ is network pressure, $W_t$ normalized waiting, $X_t$ a starvation violation, and $C_t$ an unnecessary-switch penalty. Coefficients and normalization will be fixed on training/tuning scenarios. Emergency performance is evaluated in a separate supervisory experiment so that the first normal-controller com-

parison is not confounded by rare mission rewards.

Equation 5.3 combines network pressure, normalized waiting, starvation violations, and unnecessary switching, with every component logged separately so that one favorable aggregate cannot conceal a harmful term. Coefficients and normalization are chosen on training and tuning scenarios, then frozen. Training records episode seed, demand profile, exploration schedule, optimizer settings, network checksum, reward components, software version, and candidate checkpoints. Checkpoint selection follows a declared tuning criterion rather than the best result on the final scenarios. Evaluation uses deterministic policy settings where applicable and the same safety filter, demand manifests, warm-up, and measurements as the baselines. Training return is a learning diagnostic, not an evaluation result, because it depends on reward scaling and does not directly establish travel time, fairness, or safety. Emergency priority remains a separate supervisory experiment so that rare mission rewards do not confound the first normal-controller comparison. This separation also enables an honest result when a normal controller improves average delay but produces unacceptable maximum-red behavior.

## 5.5 Dynamic Graph Routing

The routing graph represents legal vehicle movements rather than only geometric road adjacency. Nodes may be junctions or lane and edge transition points, while edges retain length, allowed vehicle class, free-flow time, current travel-time estimate, queue-discharge estimate, residual signal delay, closure state, and observation time. The exact representation must reproduce SUMO route validity, including turn restrictions and internal connections, so that an algorithm cannot return a visually plausible but undrivable path. Every cost field states its unit and source, and stale observations are rejected or replaced by a declared fallback estimate. Equation 5.4 expresses the primary cost in seconds. Link travel time, signal delay, and queue delay are separated only when their estimators do not overlap; if a measured link time already contains queuing, another queue penalty would double-count congestion. A hard closure receives infinite cost, while temporary slowdowns retain finite nonnegative estimates. The complete cost snapshot is versioned with any accepted route so that a later evaluator can reconstruct why it changed.

The primary edge cost is expressed in seconds:

$$c_e(t) = \hat{T}_e^{\text{link}}(t) + \hat{D}_e^{\text{signal}}(t) + \hat{D}_e^{\text{queue}}(t) + M_e(t), \tag{5.4}$$

where $M_e(t) = 0$ for an open edge and infinity for a hard closure. Queue delay is included only if it is not already represented in the link travel-time estimate. Every cost snapshot records source and freshness.

Dijkstra's algorithm provides the reference minimum-cost path under nonnegative weights and requires

no heuristic assumption [17]. It therefore supplies a clear correctness baseline for route validity and cost even when the selected network is small enough that search speed is not a practical limitation. A* is permitted when its heuristic $h(n)$ is a lower bound formed from straight-line distance divided by the maximum allowed free-flow speed [3]. Under nonnegative signal and queue delays, this estimate remains no greater than the true remaining time and can guide the search without losing the optimality guarantee. The implementation will compare the returned cost with Dijkstra on development graphs and will revert to Dijkstra if admissibility or consistent units cannot be demonstrated. Neither algorithm creates dynamic routing by itself: the cost snapshot, refresh interval, closure handling, current-position mapping, and stability rule determine when a path changes. Algorithm 1 consequently wraps shortest-path search with validation, no-route handling, hysteresis, route versioning, and corridor cancellation.

**Data:** Current ambulance edge $s$, destination $g$, current route $R$, cost snapshot $C_t$

**Result:** Retained or revised valid remaining route

Reject stale or negative costs; apply closures to $C_t$;

$R' \leftarrow \text{ShortestPath}(s, g, C_t)$;

**if** $R'$ *does not exist* **then**

    return NO_ROUTE and cancel future corridor requests;

**end**

Compute remaining costs $J(R)$ and $J(R')$ under the same snapshot;

**if** *current route is infeasible* **then**

    accept $R'$ immediately;

**else**

    accept $R'$ only if $J(R) - J(R') > \delta$ for persistence window $\tau$ and cooldown has elapsed;

**end**

Validate route continuity from the current position;

Atomically increment route version, cancel obsolete priority, and publish the revision;

**Algorithm 1:** Congestion-aware route selection and stable rerouting

Algorithm 1 defines route selection and revision as one auditable transaction. It first rejects stale or negative costs, applies hard closures, and searches from the ambulance's valid current position. Dijkstra provides the reference optimum for nonnegative costs, while A* may be used only with the stated admissible free-flow lower bound. A missing path produces a recorded NO_ROUTE outcome and cancellation of future corridor requests rather than a partial route. When both current and candidate paths are feasible, their remaining costs are evaluated under the same snapshot. The improvement threshold $\delta$, persistence window $\tau$, and cooldown are configured on development scenarios before held-out tests, preventing small fluctuations from causing oscillation. An infeasible current path bypasses the benefit threshold, but the replacement must still pass continuity and vehicle-class checks. Acceptance

increments the route version and atomically cancels priority that is no longer justified. Computation time, revision cause, and rejected candidates are retained so route stability can be evaluated alongside ambulance travel time.

## 5.6 Moving Green Corridor

The moving corridor operates on a bounded number of intersections ahead of the simulated ambulance instead of setting an entire route green. Activation considers estimated arrival, standing queue clearance, signal state, and downstream capacity rather than distance alone. Each request identifies the mission, immutable route version, approach movement, predicted arrival, expiry, and priority level. Before serving that movement, the coordinator completes the active phase's minimum service and applies the configured yellow and all-red clearance to conflicting movements. Passage is confirmed from the ambulance's simulated edge and lane transition, with a timeout detecting a missing or stalled vehicle. The intersection then enters recovery and either completes a bounded compensation plan or returns control at a safe phase boundary. A revised route cancels obsolete future requests before new look-ahead activation, preventing two route versions from competing for a signal. These choices follow the transition and recovery emphasis in emergency-preemption research [19] while keeping all timing values as simulation policies rather than certified field parameters.

**Data:** Priority request, ambulance state, signal state, safety configuration
**if** *request is stale, route version is obsolete, or movement is invalid* **then**
    reject and log reason;
**end**
Queue the request and estimate activation time;
At activation, finish minimum green and apply legal clearance;
Serve the ambulance movement within maximum priority duration;
**if** *passage is confirmed* **then**
    release and enter recovery;
**end**
**else if** *mission is cancelled, route changes, or timeout occurs* **then**
    cancel safely and enter recovery/fallback;
**end**
Restore normal controller ownership at a valid phase boundary and record recovery time;

**Algorithm 2:** Corridor lifecycle at one intersection

Algorithm 2 makes the intersection lifecycle explicit. A stale, expired, invalid, or obsolete request is rejected before signal ownership can change. Valid requests wait for their activation condition, com-

plete lawful clearance, and serve the ambulance movement for no longer than the configured maximum priority duration. Confirmed passage, mission cancellation, route revision, or timeout all lead through recovery rather than jumping to an arbitrary normal phase. The core scope permits one active ambulance, avoiding an untested arbitration policy for competing emergencies; multiple emergency vehicles remain future work. Ordinary-traffic harm is bounded through maximum priority duration, downstream occupancy checks, and maximum-red monitoring, and the evaluation records safety-filter interventions and recovery time. Exact limits are selected through development-scenario sensitivity analysis and are reported as simulator settings, not public-road safety values. This lifecycle allows emergency benefit and network disruption to be examined together while ensuring that cancellation is as carefully specified as successful passage.

## 5.7 Integrated Control Loop

**Data:** Validated run configuration and initialized services
**while** *run is active and before terminal simulation time* **do**
    Advance SUMO by exactly one configured step;
    Read subscribed vehicle, lane, signal, and mission state;
    Construct a timestamped traffic-state snapshot;
    Update route costs and evaluate rerouting at its configured interval;
    Update corridor requests and lifecycle states;
    **if** *corridor owns an intersection* **then**
        request its safe priority action;
    **else**
        request action from the selected normal controller;
    **end**
    Validate every requested action through the phase safety filter;
    Apply accepted actions through the TraCI adapter only;
    Record applied actions, events, and metric samples;
    Publish a non-authoritative dashboard snapshot;
**end**
Finalize run status, artifacts, and checksums;

**Algorithm 3:** Single-owner simulation and decision loop

Algorithm 3 integrates normal control, route updates, and corridor ownership under one simulation-clock writer. Each iteration advances SUMO once, reads a complete observation set, creates a timestamped snapshot, and then performs routing and corridor updates at their configured intervals. For each

intersection, the corridor or the normal controller can request an action, but both pass through the same safety filter and only the TraCI adapter applies the result. Applied actions, metric samples, and mission events are recorded before a dashboard view is published. This ordering prevents a slow client, background video job, or concurrent controller from advancing simulation time or acting on a partially updated state. A snapshot carries its effective step, production time, source, and validity interval so late data is deliberately mapped or rejected. At termination, the manifest and checksums are finalized even for stopped or failed runs. The loop therefore defines both concurrency ownership and the causal order needed to replay an experimental decision.

## 5.8 Experimental Protocol

### 5.8.1 Comparison Matrix

Table 5.1 maps each research question to a controlled comparison and the evidence required to answer it. Perception is compared with held-out annotations, normal control uses matched scenarios and seeds, routing holds the signal policy fixed, and corridor evaluation holds the route logic fixed. This staged design prevents a combined emergency result from being credited to the wrong component. The robustness row then stratifies demand, incidents, seeds, and any approved observation-noise conditions, retaining failures and no-route outcomes. Measures are selected to reveal tradeoffs: average delay is accompanied by queue extremes and fairness, ambulance time is paired with ordinary-traffic impact and recovery, and route benefit is paired with revision count and computation cost. The table is a plan rather than a results summary. No numerical improvement, accuracy, sample count, or significance is asserted until the implementation produces traceable run records under this protocol. Final result tables will preserve these question and measure definitions.

**Table 5.1: Planned experiments and primary evidence**

| Question | Controlled comparison | Primary measures |
|---|---|---|
| RQ1 perception | Detector/tracker output vs. held-out sequence annotations | Precision/recall or mAP where suitable; count MAE; identity metric only when required; queue-proxy error; throughput |
| RQ2 normal control | Fixed vs. rule-adaptive vs. selected RL under matched scenarios/seeds | Mean travel and waiting time, queue mean/max, throughput, phase changes, max-red/fairness violations |
| RQ3 routing | Static/reference route vs. dynamic route, holding signal policy fixed | Ambulance travel time, delay, stops, route revisions, computation time |
| RQ4 corridor | Same routing without vs. with corridor | Ambulance time/stops; ordinary traffic delay/queues; recovery time; safety invariant violations |
| RQ5 robustness | Demand, incident, seed, and optional observation-noise strata | Effect distributions and sensitivity; failed and no-route runs retained |

### 5.8.2 Scenario Assignment and Repetition

The evaluation manifest will define demand levels, origin-destination pairs, incident timing, dispatch time, warm-up, measurement horizon, and repeated seeds before final runs. Every controller receives the same requested set, and actual departures are retained when congestion changes insertion feasibility. The number of repetitions is chosen using pilot variance and available computation, then disclosed rather than retroactively reduced to favorable cases. Results include variability through standard deviation or confidence intervals, with paired differences used when runs share a demand seed. Reinforcement-learning training and hyperparameter selection use only training and tuning scenarios. Held-out demand combinations and at least one unseen seed set remain reserved for final evaluation, while perception data is split by source or continuous sequence to reduce temporal leakage. Failures, timeouts, infeasible routes, and safety-filter interventions remain in the run inventory and are reported separately when aggregation would be misleading. These rules preserve the distinction between incomplete implementation evidence and an evaluated negative result.

### 5.8.3 Metric Definitions

Metric definitions are shared across controllers and versioned with the run. For $N$ completed vehicles with travel times $T_i$, mean travel time is $\bar{T} = N^{-1} \sum_i T_i$. Queue length is sampled at fixed simulation intervals and reported as both time mean and maximum by approach. Throughput counts vehicles crossing declared network boundaries during the measurement interval, while the demand manifest and actual departures prevent a low completion count from being misread. Fairness includes maximum red time and the approach-level delay distribution because an aggregate mean can conceal starvation. Ambulance delay is actual travel time minus free-flow route time under the same network definition. Ordinary-traffic impact covers non-emergency vehicles during the mission and a declared recovery interval. Route computation time is measured separately from simulated travel time, and revision count identifies unstable routing. For perception counts $y_k$ and estimates $\hat{y}_k$, mean absolute error is

$$\mathrm{MAE} = \frac{1}{K} \sum_{k=1}^{K} |y_k - \hat{y}_k|. \tag{5.5}$$

### 5.8.4 Reproducibility Package

Each run will retain the source revision, dependency manifest, SUMO version, network and route checksums, controller and model version, full configuration, seed, logs, metric schema, and terminal reason. Applied decisions are stored separately from controller requests, and route revisions identify the cost snapshot and trigger that produced them. Large external video and pretrained weights may be referenced by a license-aware URI and checksum when redistribution is not permitted. A script should recreate a run from its manifest, and every manual NetEdit change must be committed as a versioned network artifact rather than left in an operator's local state. Derived plots and summary tables retain their source run identifiers and analysis-script revision. Failed and cancelled runs keep their partial logs and explicit terminal state. This package does not guarantee identical hardware timing, but it makes scenario semantics, causal events, and analytical transformations inspectable enough for another team member to reproduce the experiment or explain a difference.

## 5.9 Threats to Validity

Construct validity is threatened when a queue proxy, waiting-time implementation, or reward component does not represent the concept named in the report. Definitions and units are therefore versioned, and image-space proxies are not relabeled as physical quantities without calibration. Internal validity can be weakened by unmatched demand, different warm-up, hidden seed changes, or controller-specific

failures; paired manifests, one measurement implementation, and retention of failed runs reduce these risks. External validity remains limited by synthetic demand, map inaccuracies, SUMO car-following assumptions, a small selected network, and domain shift between public video and Lahore traffic. Conclusions will be bounded to the tested network, demand families, software versions, and video sources, with no claim about clinical response time or public-road safety. Conclusion validity can be weakened by too few repetitions, test-set tuning, selective failure removal, or reporting only means. Pilot variance, held-out scenarios, paired comparisons, variability, and sensitivity analysis address those concerns. If statistical tests are used, their assumptions and effect sizes will be reported; significance alone will not establish practical value.

## 5.10 Conclusion

The methodology specifies a staged progression from validated measurements and baselines to integrated emergency control. Network provenance and demand manifests support repeatable scenarios; source-separated video evaluation prevents perception estimates from being confused with simulator truth; and common safety constraints permit a fair controller comparison. Routing uses time-consistent nonnegative costs, validity checks, and hysteresis, while the corridor lifecycle makes request, clearance, active priority, passage, cancellation, and recovery observable. The comparison matrix separates perception, normal control, routing, corridor effect, and robustness so the cause of a result can be explained. Held-out scenarios, repeated seeds, explicit metrics, retained failures, and reproducibility artifacts protect the final evaluation from undocumented tuning. Most importantly, the procedure allows a learned controller, tracker, rerouting policy, or corridor configuration to produce a negative or mixed finding without prompting selective changes to the test protocol. The next chapter maps these methodological responsibilities to components, interfaces, state ownership, and failure handling.

# Chapter 6 High-Level and Low-Level Design

This chapter maps the requirements and methodology to a service-oriented prototype design that can be implemented incrementally on the team's available machines. The decomposition protects experimental determinism by assigning the simulation clock and all signal actuation to one adapter. The API, dashboard, perception worker, controllers, routing logic, and experiment store exchange typed messages without competing for that authority. The design also separates a requested decision from the action applied after safety checks, preserves the source and effective time of every traffic snapshot, and binds corridor requests to an immutable route version. These choices support replay and fault diagnosis without requiring a prematurely distributed deployment. Interfaces may initially be implemented as in-process modules, provided that they retain the same ownership boundaries and contracts. The chapter moves from architecture and domain objects to interaction sequences, state transitions, data contracts, synchronization, failure handling, module policies, and testing so each important requirement has a visible design consequence.

## 6.1 System Overview and Design Goals

The system must support repeatable research runs, an interactive demonstration, and post-run analysis without allowing presentation concerns to alter experimental state. Its main design goals are deterministic ownership of state changes, replaceable controller and perception implementations, explicit time and units, safe corridor transitions, immutable run configuration, and visible failure. Performance is optimized only after these properties are preserved because a faster result is not useful when its source, phase legality, or causal order is uncertain. The design assumes one machine or a small development deployment for Deliverable 2 and FYP-I. Logical separation therefore does not imply a network service for every box in the diagrams. The orchestrator, controllers, routing logic, and repositories may begin as in-process modules behind typed interfaces, while video inference can move to a bounded worker when profiling shows it would block the control loop. This approach keeps deployment manageable but preserves a path to isolate resource-heavy components without changing experimental semantics.

## 6.2 High-Level Architecture

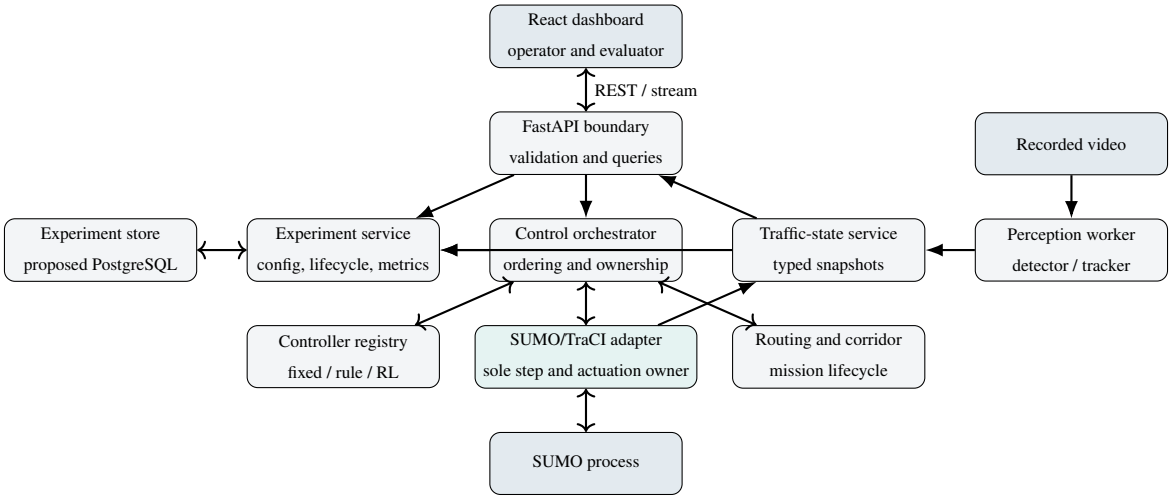



**Figure 6.1: Proposed high-level architecture**

Figure 6.1 shows the command and evidence paths around the single-owner simulation core. The React dashboard sends configuration or mission commands through FastAPI and receives non-authoritative views; it never calls TraCI. The orchestrator follows Algorithm 3, requests decisions from the selected normal controller or corridor coordinator, and sends validated actions to the adapter. Only that adapter advances SUMO and applies signal changes. The traffic-state service constructs source-specific snapshots from simulator observations or perception messages while preserving effective time, units, confidence, and validity. Routing maintains mission and route versions, and the corridor coordinator owns intersection-level priority lifecycles. The experiment service validates run configuration, records events and metrics, and persists them through a repository interface whose proposed implementation is PostgreSQL. Directional arrows in the figure distinguish state flow from command ownership, making clear that perception can inform a snapshot without acquiring authority over simulation time. This separation also keeps dashboard latency outside the control path.

## 6.3 Component Responsibilities and Interfaces

Table 6.1 assigns each component the data it consumes and the state or evidence it is permitted to produce. This allocation prevents shared mutable state from becoming an undocumented integration mechanism. The dashboard owns only presentation state, controllers own only proposed actions and explanations, and the SUMO adapter owns the external process, subscriptions, clock, and applied actuation. The orchestrator orders these modules but does not replace their domain logic. Routing and corridor responsibilities are listed separately because a shortest path does not imply authority to change

a signal, and a corridor request must retain the route version that justified it. The traffic-state service can normalize observations from two source modes but may not silently combine missing video measures with exact simulator values. Finally, the experiment store receives validated records and artifacts rather than arbitrary objects from every module. These boundaries give each requirement a testable owner and permit an implementation to remain modular even when several components share one process.

**Table 6.1: Component responsibility allocation**

| Component | Consumes | Produces / owns |
|---|---|---|
| React dashboard | Query responses, event snapshots, user input | Presentation state; no control authority |
| FastAPI boundary | HTTP/stream commands, identity-free local operator session | Validation, command/query boundary, error mapping |
| Experiment service | Valid configuration, metric/event batches | Run lifecycle, immutable manifest, comparison eligibility |
| Control orchestrator | Run config and ordered state snapshot | Decision schedule, module ordering, requested actions |
| SUMO/TraCI adapter | Validated step and action commands | Simulation process, clock, subscriptions, applied actions |
| Traffic-state service | SUMO observations or perception messages | Versioned source-specific snapshots and freshness |
| Perception worker | Video and model configuration | Detections, tracks, aggregates, job health |
| Controller registry | State, legal phases, timing constraints | Requested normal-control action and explanation |
| Routing service | Graph, mission, cost snapshot | Route versions, no-route reason, reroute decision |
| Corridor coordinator | Route version, ambulance state, signal state | Priority request/state, release and recovery requests |
| Experiment store | Validated records and artifacts | Durable configuration, events, metrics, checksums |

## 6.4 Architectural Decisions and Tradeoffs

Table 6.2 records the main alternatives considered and the consequence of each selected approach. The choices favor traceability and feasible integration over maximum distribution or algorithmic novelty. SUMO remains the simulator because it aligns with the approved proposal and provides detailed route and signal inspection, although CityFlow could offer greater training throughput for a different scope. A common controller interface and safety filter make fixed, rule-based, and learned policies comparable, while supervisory emergency control permits a separate corridor ablation. Dijkstra remains the routing reference and A* is conditional on an admissible time lower bound. Structured persistence is proposed because flat files make cross-run relationships difficult to enforce, but the repository boundary keeps the engine replaceable. The decision record is not evidence that every selected technology has been completed; it explains why the design adopts these constraints and what cost each creates, including additional message schemas, job states, and performance profiling.

**Table 6.2: Architectural decision record**

| **Decision** | **Selected approach** | **Alternative and reason not selected initially** | **Consequence** |
|---|---|---|---|
| Simulation ownership | One orchestrator/TraCI adapter controls stepping and signals | Multiple services calling TraCI risks nondeterministic order | Simple replay and concurrency model |
| Simulator | SUMO, as approved and supported by detailed road/signal tooling | CityFlow offers training throughput but changes the approved environment | Lower migration risk; performance profiled later |
| API/UI | FastAPI plus React, fixed by proposal | Server-rendered monolith would simplify deployment but reduce clear service contracts | Typed API and independent UI iteration |
| Controller integration | Common request interface plus safety filter | Direct controller access to TraCI couples research code to simulation | Comparable controllers; applied action may differ from request |

| Decision | Selected approach | Alternative and reason not selected initially | Consequence |
|---|---|---|---|
| Emergency control | Supervisory corridor state machine | Adding emergency reward directly to normal RL confounds attribution | Clear ablation and safety ownership |
| Routing | Dijkstra reference; A* only with admissible lower bound | Distance-only route is simpler but ignores the research question | Time-based, auditable costs |
| Persistence | Structured experiment schema; PostgreSQL proposed | Flat files are quick but weak for cross-run integrity | Engine remains replaceable behind repository interface |
| Perception execution | Background worker with bounded messages | In-request inference blocks API and control loop | Adds job state and queue management |

## 6.5 Domain Model and Class Diagram

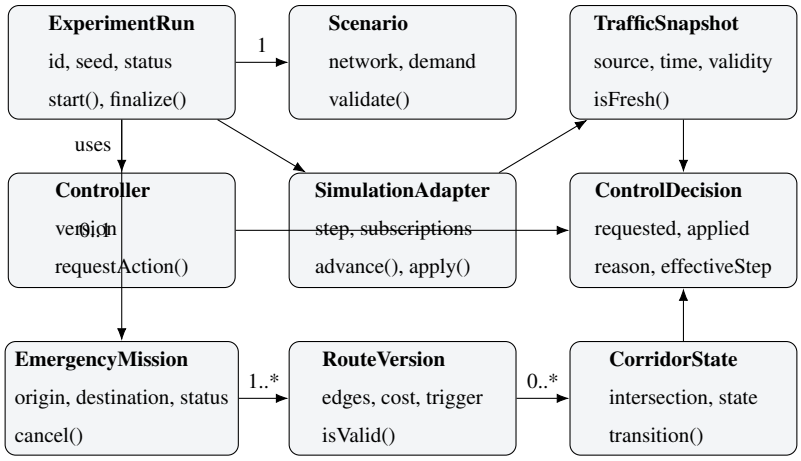



**Figure 6.2: Proposed domain model/class diagram**

Figure 6.2 presents the core domain objects and their relationships independently of a particular web or database framework. An ExperimentRun binds one validated Scenario to a controller, simulation adapter, generated snapshots, decisions, and an optional emergency mission. The model deliberately

separates a ControlDecision's requested phase from its applied phase. A controller can request an early or otherwise invalid switch, but the safety filter and adapter determine what actually reaches SUMO and record the intervention reason. TrafficSnapshot also carries source, simulation time, and validity so a decision can be traced to the observation it used. EmergencyMission refers to one current immutable RouteVersion, while previous versions remain available for analysis. CorridorState objects refer to the version that justified their request, enabling the coordinator to reject or cancel obsolete priority after rerouting. The diagram represents proposed classes and responsibilities; exact method signatures may change during implementation without weakening these invariants.

## 6.6 Interaction Sequences

### 6.6.1 Normal Control Update

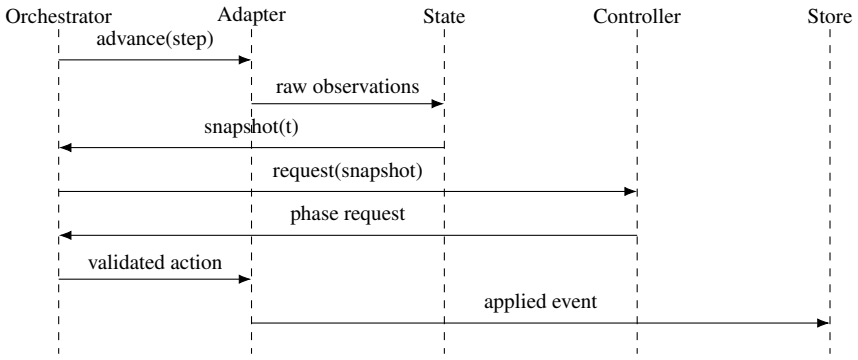



**Figure 6.3: Sequence for a normal signal-control update**

Figure 6.3 shows the causal order for one normal-control decision. The orchestrator first asks the adapter to advance one simulation step, after which raw subscriptions are normalized into a complete snapshot. Only then does the selected controller receive state and return a phase request. The orchestrator validates that request against phase membership, elapsed minimum green, clearance, and ownership before the adapter applies it, and the resulting event is sent to the store. A controller therefore never observes a partial update and cannot actuate SUMO as a side effect. When a controller decision interval spans several simulation steps, the last applied action remains in force while state and metrics continue to be sampled at their declared intervals. The stored event includes both requested and applied action, effective step, snapshot step, and intervention reason. This sequence allows a replay to distinguish a controller choice from a safety transformation and prevents dashboard timing from entering the decision path.

### 6.6.2 Ambulance Dispatch and Corridor Activation

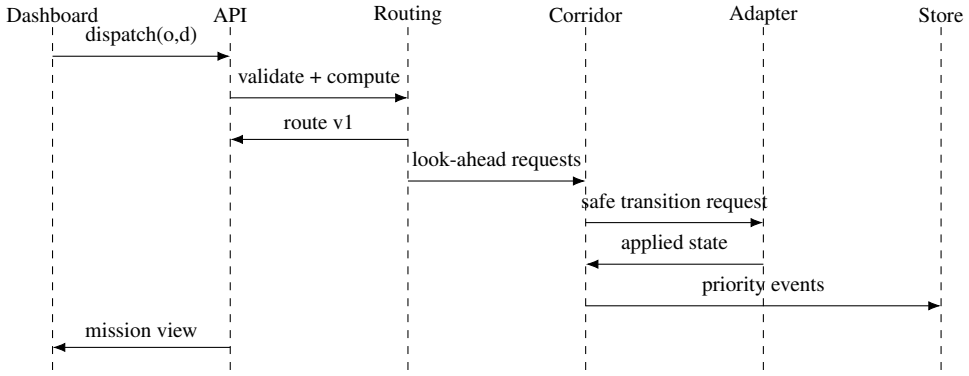



**Figure 6.4: Sequence for ambulance dispatch and corridor activation**

Figure 6.4 begins when the operator submits an origin and destination through the dashboard. FastAPI validates the command and asks the routing service to resolve accessible edges and compute the first immutable route version. A successful route is returned for the mission view and supplied to the corridor coordinator, which creates bounded look-ahead requests rather than activating the whole path. Each due request passes through safe transition before the adapter applies priority, and the corridor writes state changes and passage evidence to the experiment store. Invalid endpoints or a disconnected graph stop the sequence before an ambulance or partial corridor is activated. The dashboard receives a derived mission view after the authoritative state has changed, so disconnecting the client cannot cancel or delay the mission. This sequence separates route computation, signal ownership, and presentation while retaining the identifiers required to trace an active priority state back to the operator command and route version.

### 6.6.3 Rerouting and Recovery

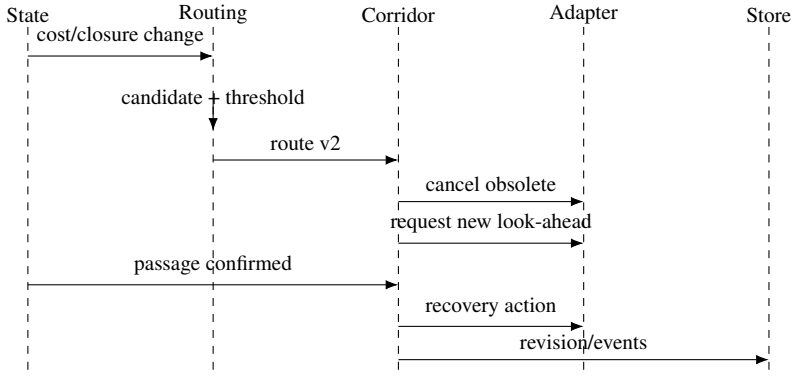



**Figure 6.5: Sequence for route revision, obsolete-priority cancellation, and recovery**

Figure 6.5 shows how a closure or sustained cost change is handled after dispatch. The routing service evaluates a candidate from the ambulance's valid current position and applies the configured feasibility,

benefit, persistence, and cooldown rules. If accepted, a new immutable route version is published to the corridor coordinator. Requests associated with the old remaining path are cancelled before the new look-ahead set is activated, preventing simultaneous priority for incompatible route versions. Passage confirmation still originates from traffic state and moves the relevant intersection into recovery, while route revisions, cancellations, requests, and recovery events are stored in causal order. When no feasible replacement exists, the mission becomes blocked and upcoming priority is cancelled rather than preserving a corridor that no longer reaches the destination. This interaction makes rerouting an auditable state transition, not merely a line redrawn on the dashboard. Rejected candidates also remain available for diagnosis.

## 6.7 Corridor State Model

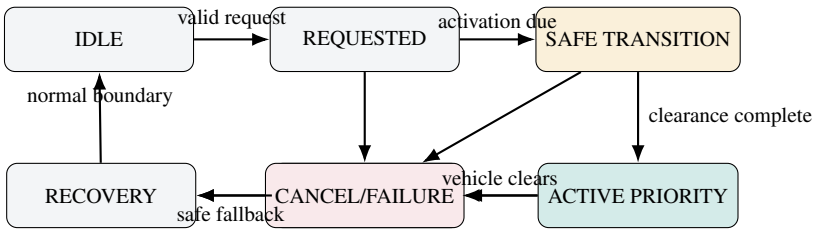



**Figure 6.6: Intersection-level moving-corridor state transitions**

Figure 6.6 defines the lifecycle owned by each affected intersection. IDLE accepts a valid route-versioned request but REQUESTED does not yet imply signal ownership. When activation becomes due, SAFE TRANSITION waits for the current phase's minimum service and completes the configured yellow and all-red clearance. ACTIVE PRIORITY serves only the ambulance movement and is bounded by a maximum duration. Confirmed vehicle clearance creates a PASSAGE event, after which RECOVERY restores a safe normal boundary before the ordinary controller regains direct ownership. Cancellation, timeout, invalidation, or route revision can occur while a request is queued, transitioning, or active, but none of these paths jumps directly to an arbitrary phase. Instead, the failure branch enters the same guarded recovery process. This state model makes both successful passage and interrupted missions testable through transition guards, event records, ownership assertions, and eventual return to IDLE. Every transition records its trigger and effective step.

## 6.8 Traffic-State and Event Contracts

Table 6.3 lists the minimum fields carried across important module boundaries. Version and time fields prevent a consumer from accepting an unknown schema or treating an old snapshot as current. Source

mode, units, confidence, and validity distinguish simulator values from calibrated or proxy video observations. SignalState identifies both the legal successors and current owner, while ControlRequest has no direct side effect and can be transformed into an authoritative AppliedControl record by the safety path. RoutePlan and PriorityRequest carry the mission and route version needed to reject obsolete corridor work. MetricSample retains its definition version and interval so summaries cannot combine measurements whose meaning changed during development. Field names are conceptual and may be mapped to Python or JSON naming conventions, but their semantics remain stable. This contract-first approach also makes integration tests possible before every research component is complete, because a stub can produce valid messages without bypassing ownership rules.

**Table 6.3: Core interface and data-contract fields**

| Contract | Required fields | Rule |
|---|---|---|
| TrafficSnapshot | schemaVersion, sourceMode, producedAt, effectiveStep, validUntil, networkId | Consumers reject unknown schema or expired validity; wall time and simulation time are distinct. |
| ApproachState | intersectionId, approachId, count, queue, wait, speed, occupancy, units, confidence | Missing is not zero; uncalibrated physical quantities are omitted. |
| SignalState | intersectionId, programId, phaseId, phaseElapsed, legalNext, owner | Owner is NORMAL or CORRIDOR; legal next phases come from validated configuration. |
| ControlRequest | controllerId, intersectionId, requestedPhase, duration/extension, snapshotStep, reason | Request has no side effect and may be rejected or transformed. |

| Contract | Required fields | Rule |
|---|---|---|
| AppliedControl | requestId, appliedPhase, effectiveStep, safetyIntervention, reason | This record is authoritative for analysis. |
| RoutePlan | missionId, routeVersion, edgeIds, costSeconds, snapshotStep, trigger | Edge order must be continuous from valid current position. |
| PriorityRequest | missionId, routeVersion, intersectionId, movementId, ETA, expiresAt | Obsolete route version or expired request is rejected. |
| MetricSample | runId, metricName, metricVersion, interval, value, unit | Aggregation across mismatched versions is prohibited. |

Event ordering is based on simulation step within a run and a monotonic sequence number assigned by the orchestrator. Wall-clock timestamps support diagnostics but do not determine traffic causality because inference, rendering, and storage latency can vary independently of simulated time. A snapshot identifies the step at which its values became effective, and an applied signal event records the step at which the adapter enacted it. Route revisions and priority events are enqueued in an order that prevents an old request from being interpreted as current. Missing values remain missing rather than being converted to zero, and unknown schema versions are rejected rather than partially decoded. These rules give the fields in Table 6.3 operational meaning and support replay, stale-data handling, and comparison across runs without assuming perfectly synchronized wall clocks. They also allow a delayed write to be distinguished from a delayed decision.

## 6.9 Concurrency, Synchronization, and Resource Ownership

The orchestrator is the only writer to simulation time, and TraCI access is serialized behind the adapter. Controller computation receives an immutable snapshot and returns a request rather than holding a reference to mutable global state. Perception uses a bounded work queue; if inference falls behind, the worker marks observations late or drops work according to a recorded policy instead of blocking control

indefinitely. Dashboard streams are lossy views, so a slow client may skip intermediate snapshots but cannot delay decisions. Run configuration becomes immutable after start. The current route version is updated atomically with cancellation of obsolete corridor requests, preventing new and old paths from owning priority concurrently. Database writes may be batched where safe, but an applied signal event or mission transition is not acknowledged until it has entered a durable bounded queue. During shutdown, the API stops accepting new commands, the orchestrator ends or detaches SUMO safely, queued records are flushed, and the run receives an explicit completed, stopped, failed, or incomplete terminal state.

## 6.10 Error Handling and Fallback

Table 6.4 defines immediate containment and retained evidence for failures at each major boundary. A SUMO or TraCI failure stops decision-making because continuing to show live state would be misleading, whereas a dashboard disconnect leaves the authoritative experiment unaffected. Stale observations and invalid controller requests are rejected or handled through a declared simulator or fixed-control fallback, and every interval using that fallback is marked so it cannot be included silently in a clean learned-controller aggregate. A missing route prevents partial corridor activation, while a lost ambulance or timeout cancels future priority and moves active intersections through recovery. Persistence uses a bounded buffer only long enough to tolerate a transient issue; the run stops before evidence loss grows without limit and is marked incomplete if durability cannot be restored. These policies favor an explicit failed run over an apparently successful record whose causal events are missing. Fault-injection and small-network integration tests will verify both the immediate behavior and the final recorded state.

**Table 6.4: Failure handling strategy**

| Failure | Immediate behavior | Recorded/recovery behavior |
|---|---|---|
| SUMO launch/TraCI disconnect | Stop decisions and do not present live state | Mark run failed; retain command, stderr, step, and manifest |
| Stale observation | Exclude or use declared simulator fallback | Record staleness and selected fallback source |
| Invalid controller action | Reject through safety filter | Apply fixed/safe continuation and record intervention |
| RL service unavailable | Switch at valid boundary to configured baseline | Record fallback interval; exclude from clean RL aggregate or report separately |
| No route | Do not spawn/activate partial corridor | Mark mission blocked; show cause and graph snapshot ID |
| Lost ambulance/timeout | Cancel future priority and recover active intersections | Mark mission failed/cancelled and preserve last position |
| Persistence failure | Buffer within a bounded queue; stop before unbounded evidence loss | Mark run incomplete if durability cannot be restored |
| Dashboard disconnect | Continue experiment unaffected | Reconnect from current snapshot and event cursor |

## 6.11 Detailed Module Policies

### 6.11.1 Controller Policy

All controllers implement a common method that accepts an immutable traffic snapshot, intersection configuration, and decision context, then returns a request with explanation fields. They cannot write global state, call TraCI, or suppress the record of a rejected action. The fixed controller, pressure-inspired controller, and learned policy therefore differ in decision logic while sharing phase identifiers, decision intervals, safety constraints, and applied-event recording. Deterministic evaluation disables exploration and seeds every permitted stochastic operation. The safety filter verifies phase membership, minimum green, required clearance, maximum priority duration, and current ownership before an action reaches the adapter. If a policy is unavailable or repeatedly invalid, control transfers at a legal boundary to the configured baseline and the affected interval is marked. This common interface is central to a

fair comparison because it prevents one controller from receiving more permissive movements or direct simulator privileges. Explanation fields are retained for later error analysis.

### 6.11.2 Routing Policy

The graph builder derives nodes and edges from validated SUMO vehicle movements, including allowed classes and turn continuity. Cost providers are composable but must state units, observation time, fallback behavior, and whether link travel time already includes queue delay. This metadata prevents the routing service from adding overlapping estimates or comparing seconds with an unnormalized congestion score. A published route plan is immutable, and the mission points to exactly one current version. Revisions are accepted only through Algorithm 1, which compares paths under one cost snapshot and applies feasibility, hysteresis, persistence, and cooldown rules. A hard closure overrides ordinary cost and invalidates an affected remaining path immediately. Before publication, the route is checked from the ambulance's current edge to the destination and given a revision cause. The routing service never actuates signals; it supplies versioned plans to the corridor coordinator, which can then cancel or create priority requests consistently.

### 6.11.3 Persistence Policy

The store uses schema migrations and referential constraints for structured configuration, events, missions, routes, and metrics, while large videos, model files, and logs may remain checksummed external artifacts with provenance. Raw high-frequency state can be downsampled for persistence only when the sampling interval and aggregation rule are recorded in the metric definition. Completed run configuration is immutable, and derived summaries retain their source run identifiers and analysis version. Secrets, machine-specific credentials, and unportable local paths are excluded from research exports; licensed artifacts are referenced rather than redistributed when necessary. PostgreSQL is proposed for structured records, but implementation may begin with a file-backed engine behind the same repository interface. The domain schema and integrity rules remain independent of that choice so a storage migration cannot change the meaning of an experiment. A persistence failure follows the bounded behavior in Table 6.4 rather than silently discarding events. Recovery records state whether buffered evidence was restored.

### 6.11.4 User-Interface Policy

The user interface is a view and command boundary, not a second control plane. Colors are supplemented with text and symbols so congestion, corridor state, or failure is not conveyed by color alone. Values display their units, source mode, freshness, and invalid state where applicable. Schematic predic-

tions remain visually distinct from measured results, and an uncalibrated video proxy is not labeled as a physical queue or speed. The interface does not display "time saved" until a valid reference run exists under the same comparison manifest, and it exposes configuration differences when direct comparison is unsafe. Commands receive validation errors tied to fields, while connection loss leaves the experiment running and allows a client to reconnect from the current event cursor. These policies implement the wireframe intentions in Section 4.7 and ensure that a polished display cannot conceal stale data, fallback intervals, missing runs, or incomplete evidence.

## 6.12 Testing Strategy

Unit tests cover cost functions, route continuity, hysteresis, metric calculations, schema validation, freshness rules, and corridor transition guards. Property or invariant tests generate action sequences to ensure incompatible greens are never requested for application, one component owns each intersection, obsolete route versions cannot activate priority, and cancellation eventually reaches IDLE through RECOVERY. Integration tests use small SUMO networks to validate single-step ownership, detector-to-state messages, route application, fallback, and persistence ordering. Scenario tests replay fixed seeds for normal control, closure rerouting, no-route, invalid actions, stale data, controller loss, ambulance timeout, persistence failure, and dashboard disconnect. Contract fixtures allow components to be replaced by stubs without weakening field validation. Research verification remains separate from software correctness: passing these tests shows that the implementation follows its specification, but it does not establish that a policy improves traffic or emergency travel. The matched experiments in Chapter 5 provide that evidence, while test results explain whether a failed experiment reflects the policy or a software defect.

## 6.13 Coding, Build, and Maintainability Policies

Python modules will use static type hints, consistent formatting and linting, bounded dependency versions, structured logging, and documentation for public interfaces. React components separate data acquisition from presentation so a visual revision cannot change experiment ownership. Configuration is stored in validated versioned files rather than scattered constants, and secrets remain outside exported manifests. Continuous checks should run unit tests, schema validation, and a minimal headless SUMO scenario before a revision is used for evidence. Experiment releases are tagged and their commit identifier is written into each run manifest. The intended repository structure separates application services, controllers, routing, perception, simulation, dashboard, contracts, scenarios, and experiment artifacts. This is a proposed organization, not evidence that all directories or services already exist. Small interfaces allow a tracker, learned policy, database engine, or dashboard transport to change without rewriting

the central loop, while versioned contracts force such changes to declare any effect on recorded evidence.

## 6.14 Conclusion

The design assigns one clear owner to every mutable resource and makes observation source, simulation time, units, route version, requested action, and applied action explicit. The high-level architecture keeps presentation and perception outside the single-writer simulation path, while component responsibilities and data contracts make integration boundaries testable. The domain model and sequences preserve causal links from scenario and snapshot through controller request, safety decision, route revision, corridor transition, and durable event. Guarded recovery and bounded fallbacks ensure that a failure does not leave an unexplained signal or apparently live dashboard. Common controller interfaces support the intended progression from fixed and pressure-inspired baselines to a learned policy, and versioned routing enables a moving corridor without granting the pathfinder direct signal authority. The architecture is modular enough for team implementation on one machine and leaves distribution as a profiling decision, not an assumption. It also preserves failed runs and negative research outcomes as evidence instead of hiding them behind interface behavior.

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# Chapter A Deliverable 2 Requirements Coverage

Table A.1 provides a final audit trail from the governing Deliverable 2 plan to the chapters and evidence in this report. It is included for review convenience and does not replace the detailed requirements traceability matrix in Chapter 4.

**Table A.1: Requirements coverage matrix**

| Deliverable requirement | Primary evidence | Report location |
|---|---|---|
| Project purpose and vision | Approved proposal and maintained context | Chapters 1–2 |
| At least 15 relevant studies | Verified peer-reviewed literature and official papers | Chapter 3 and bibliography |
| Software requirements and use cases | Proposal scope and planned operator workflow | Chapter 4 |
| GUI and data design | Proposed dashboard and persistence needs | Sections 4.7–4.8 |
| Research methodology | Proposal baselines and reproducible evaluation plan | Chapter 5 |
| High- and low-level design | Service responsibilities and control safety constraints | Chapter 6 |
| Architecture, UML, sequences, state model | Original TikZ diagrams | Sections 6.2–6.7 |
| Honest project status | Proposal stage statement and repository evidence | Throughout; Section 1.3 |

# Chapter B Open Items for Team Confirmation

The report remains usable without resolving the items below, but they must be confirmed before an implementation or final-results report makes corresponding claims.

1. Exact Lahore road area, network size, and imported OpenStreetMap snapshot.
2. Final detector checkpoint, tracker configuration, annotated video sources, and license records.
3. Final RL algorithm and hyperparameters after baseline feasibility tests.
4. Database selection, demonstration hardware, and measured update/latency targets.
5. Actual application screenshots, once the interfaces exist; all current interface figures are labelled as proposed wireframes.
6. Any faculty-issued certificate or approval page not present in the supplied FAST template.